

BFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Complete



Processing



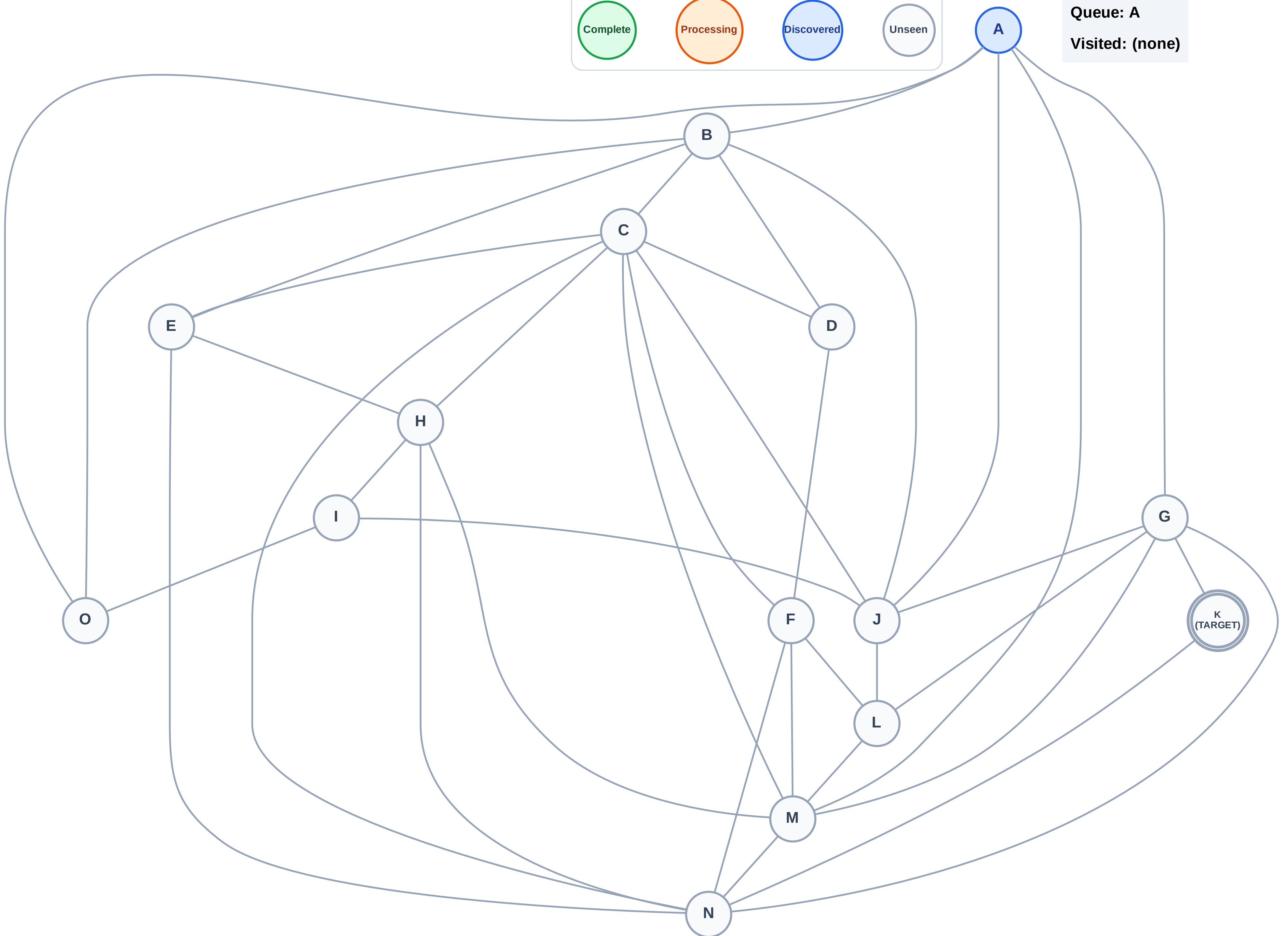
Discovered



Unseen

Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

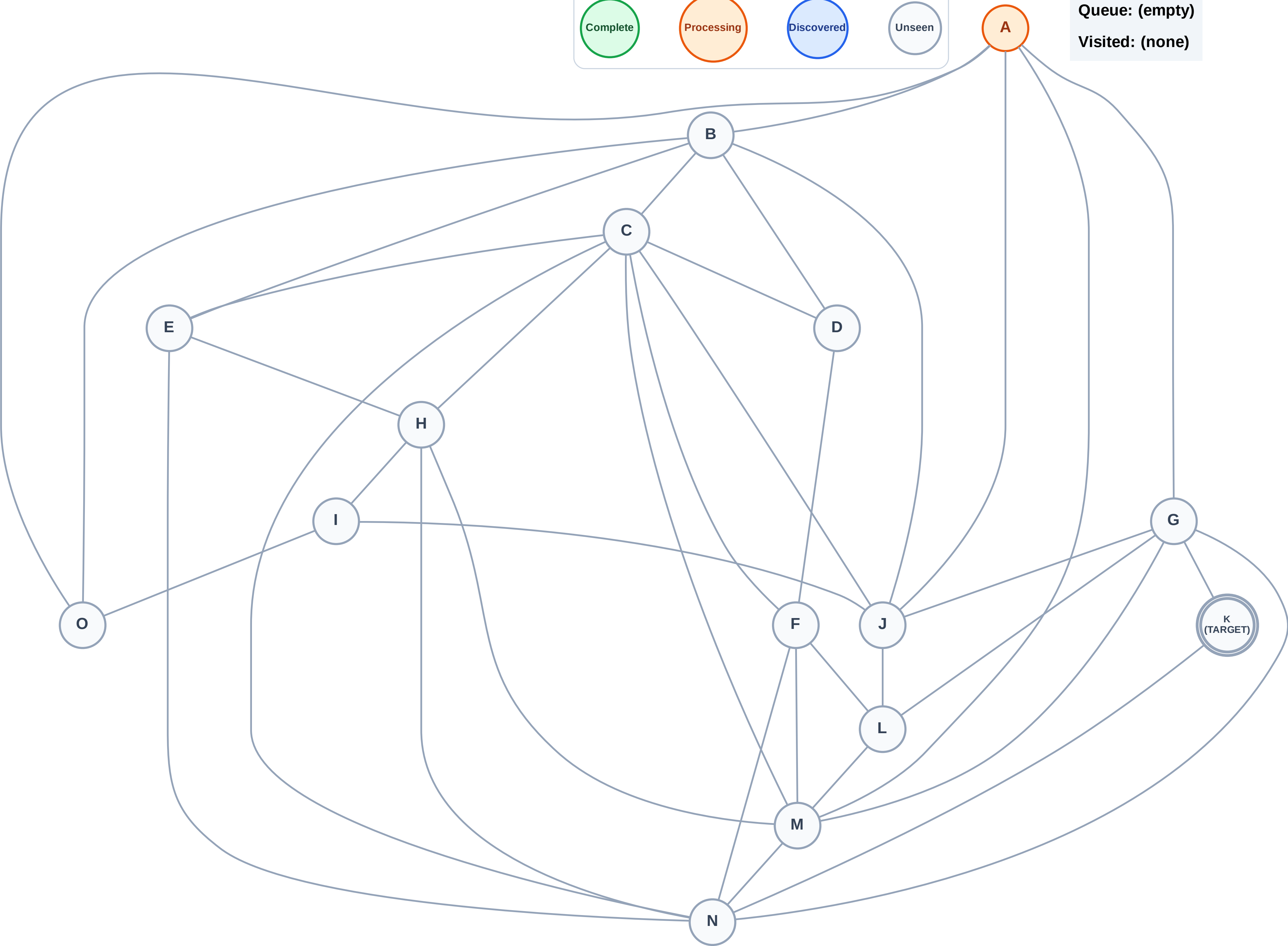
Complete

Processing

Discovered

Unseen

Queue: (empty)
Visited: (none)



BFS Traversal

Step 3: Discover B, G, J, M, O from A

Legend



Complete



Processing



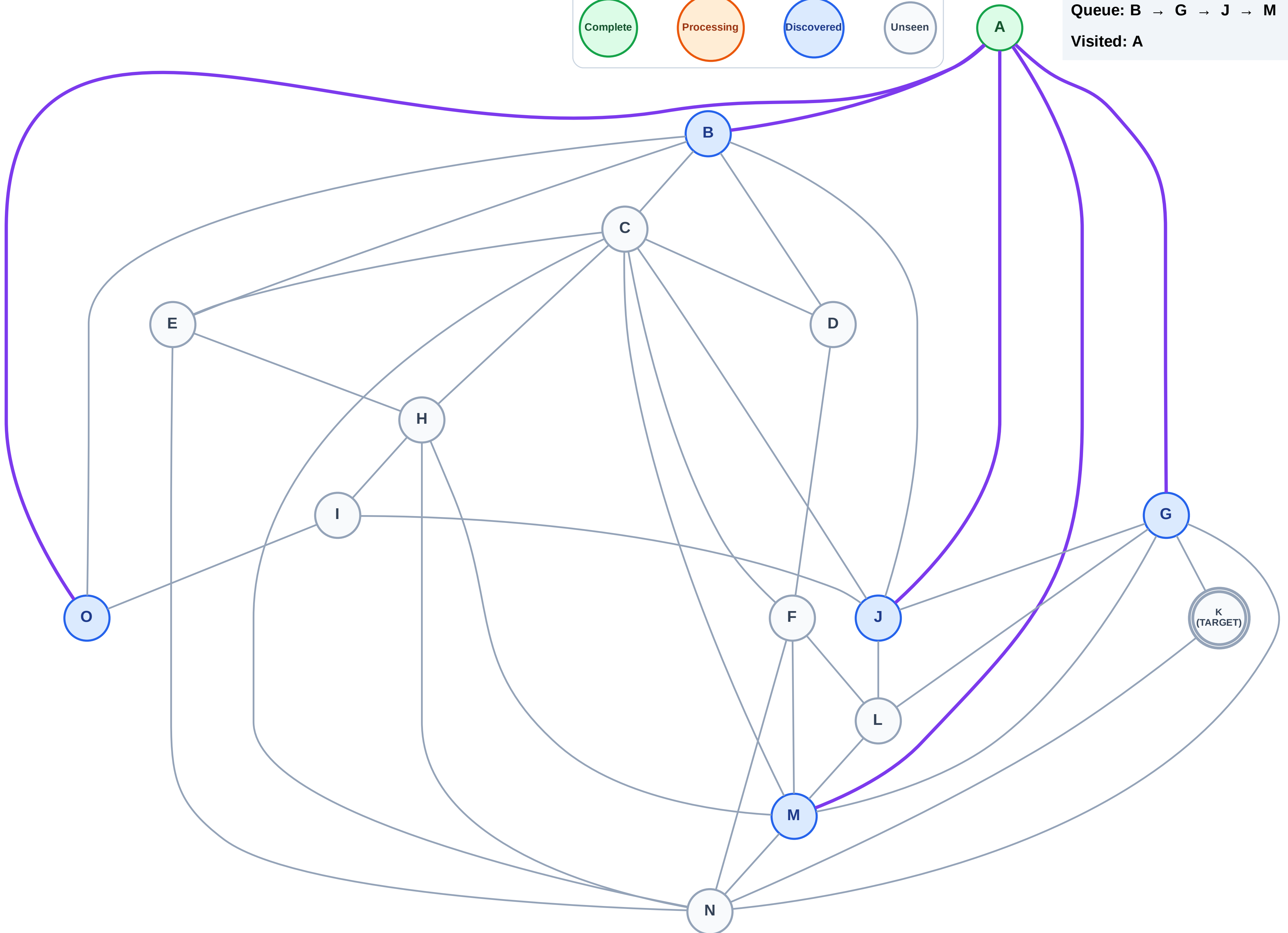
Discovered



Unseen

Queue: B → G → J → M → O

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

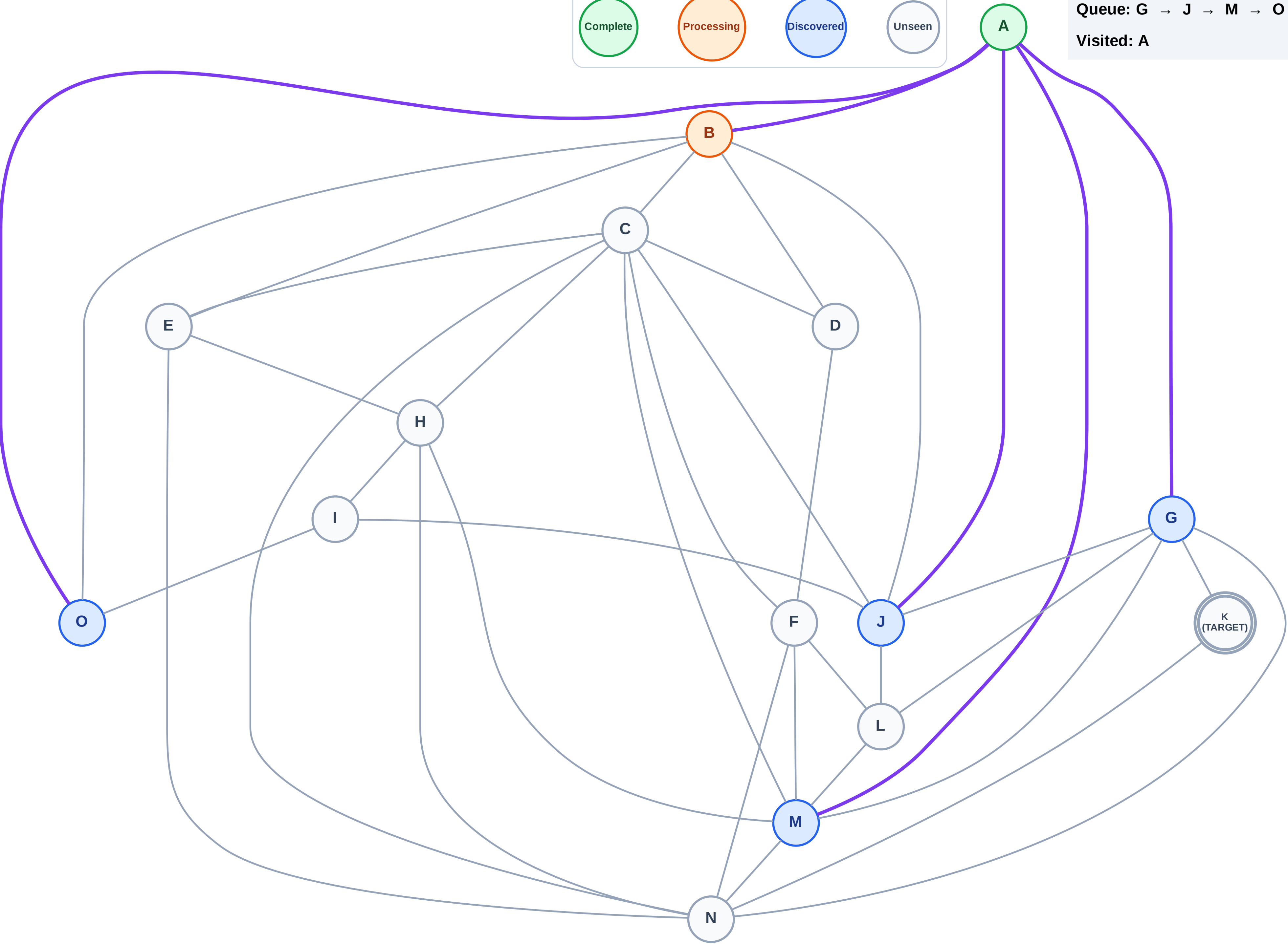
Processing

Discovered

Unseen

Queue: G → J → M → O

Visited: A



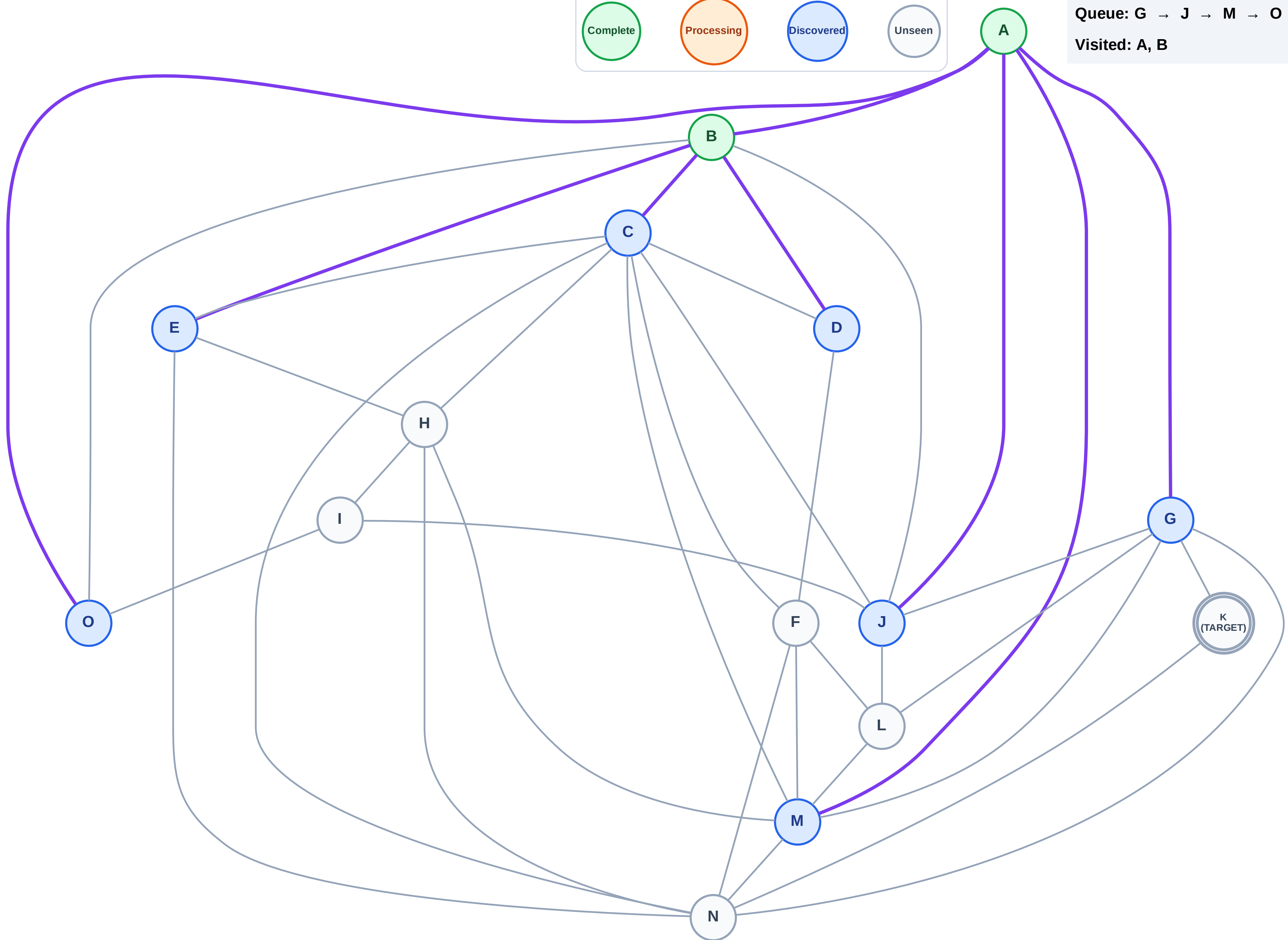
BFS Traversal

Step 5: Discover C, D, E from B

Legend



Queue: G → J → M → O → C → D → E
Visited: A, B



BFS Traversal

Step 6: Process node G

Legend



Complete



Processing



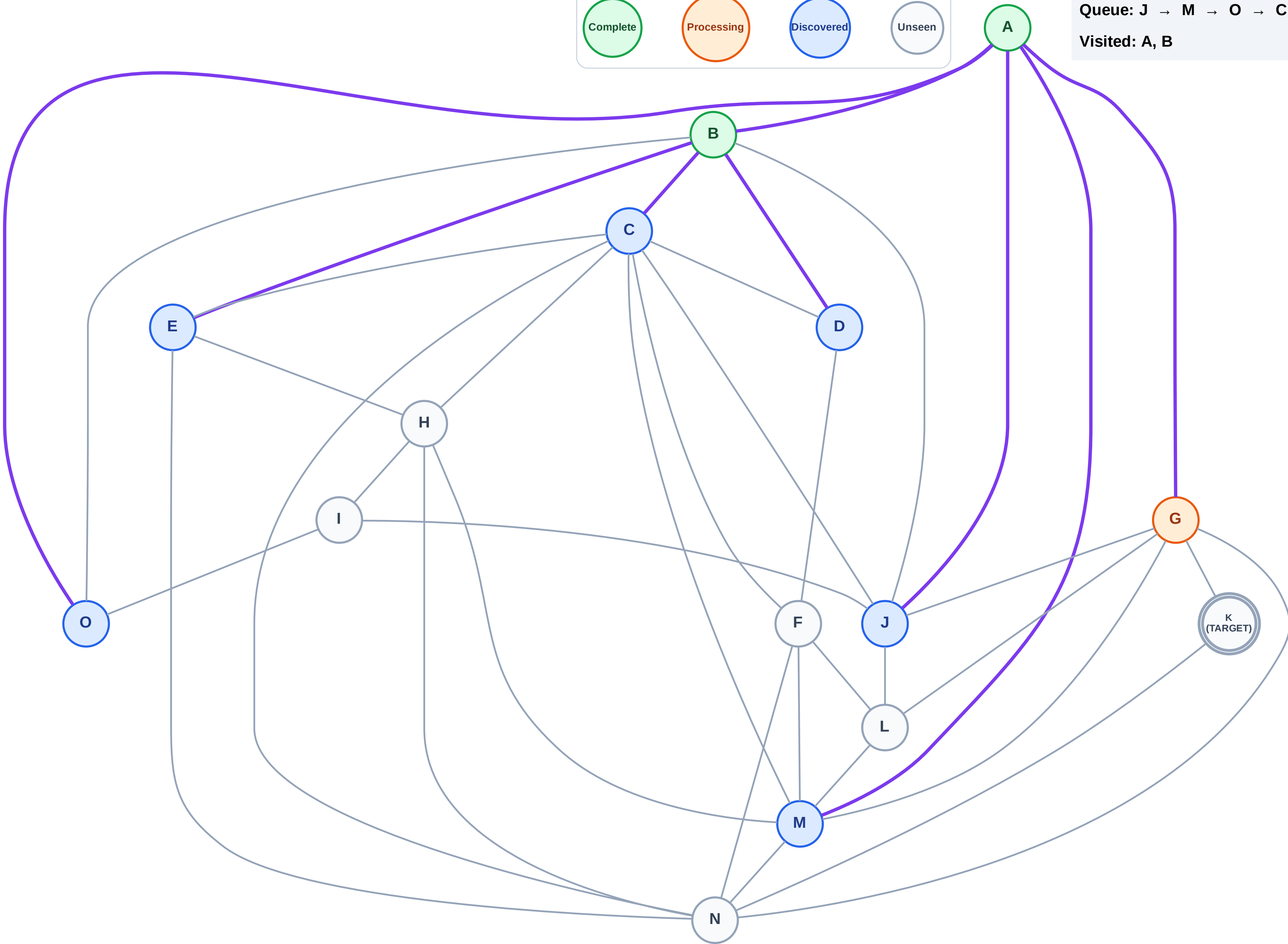
Discovered



Unseen

Queue: J → M → O → C → D → E

Visited: A, B



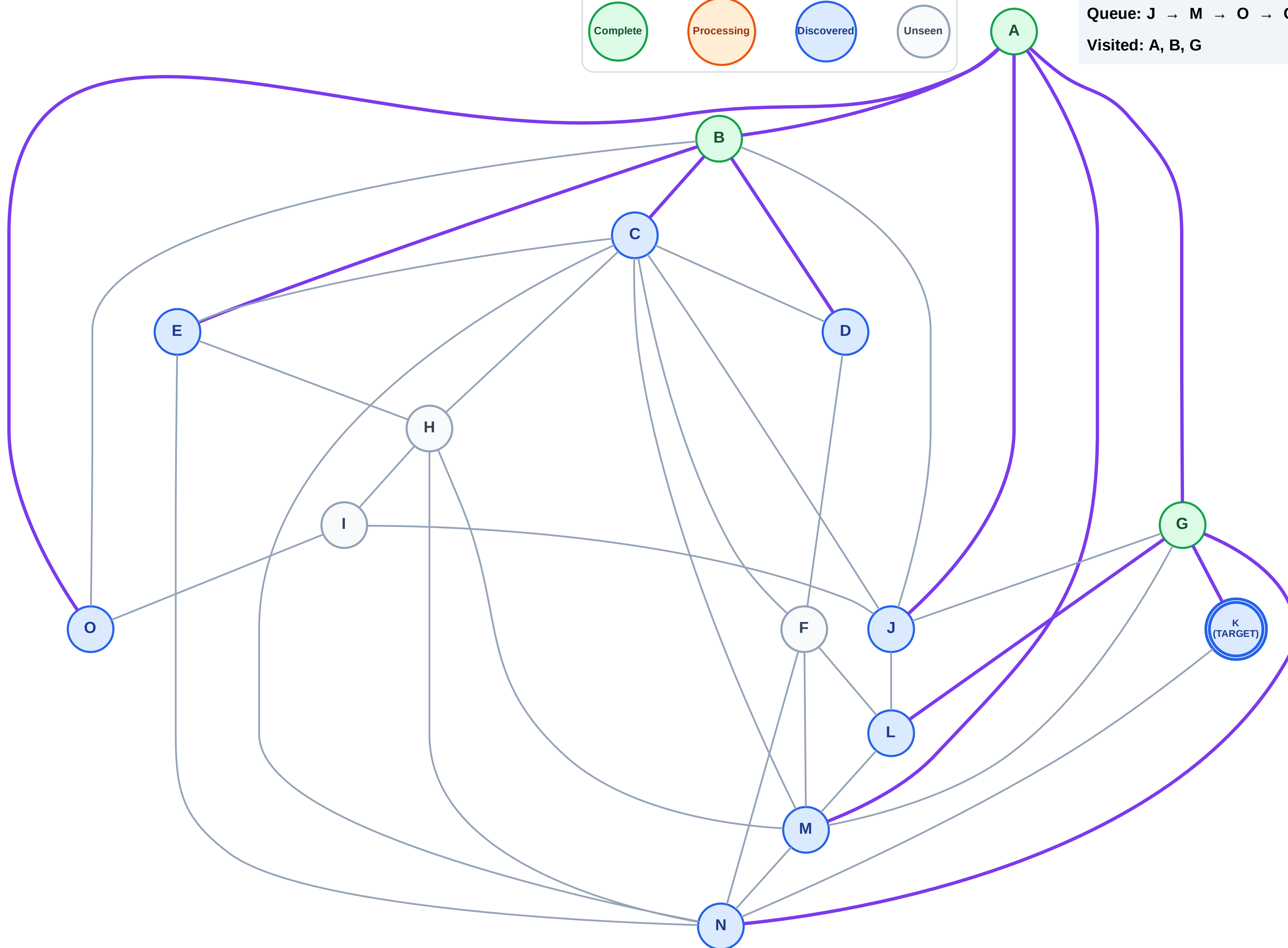
BFS Traversal

Step 7: Discover K, L, N from G

Legend



Queue: J → M → O → C → D → E → K → L → N
Visited: A, B, G



BFS Traversal

Step 8: Process node J

Legend

Complete

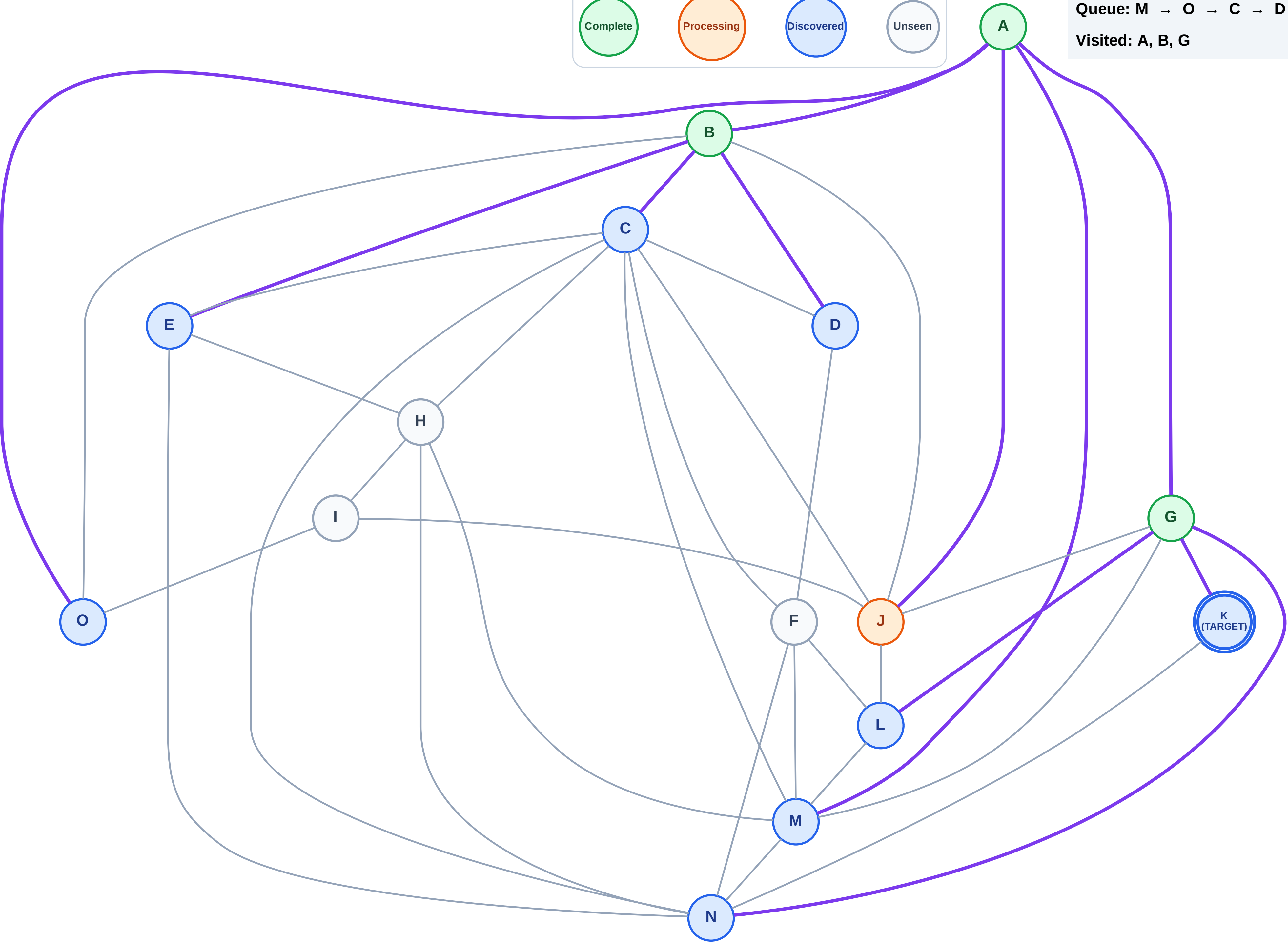
Processing

Discovered

Unseen

Queue: M → O → C → D → E → K → L → N

Visited: A, B, G



BFS Traversal

Step 9: Discover I from J

Legend

Complete

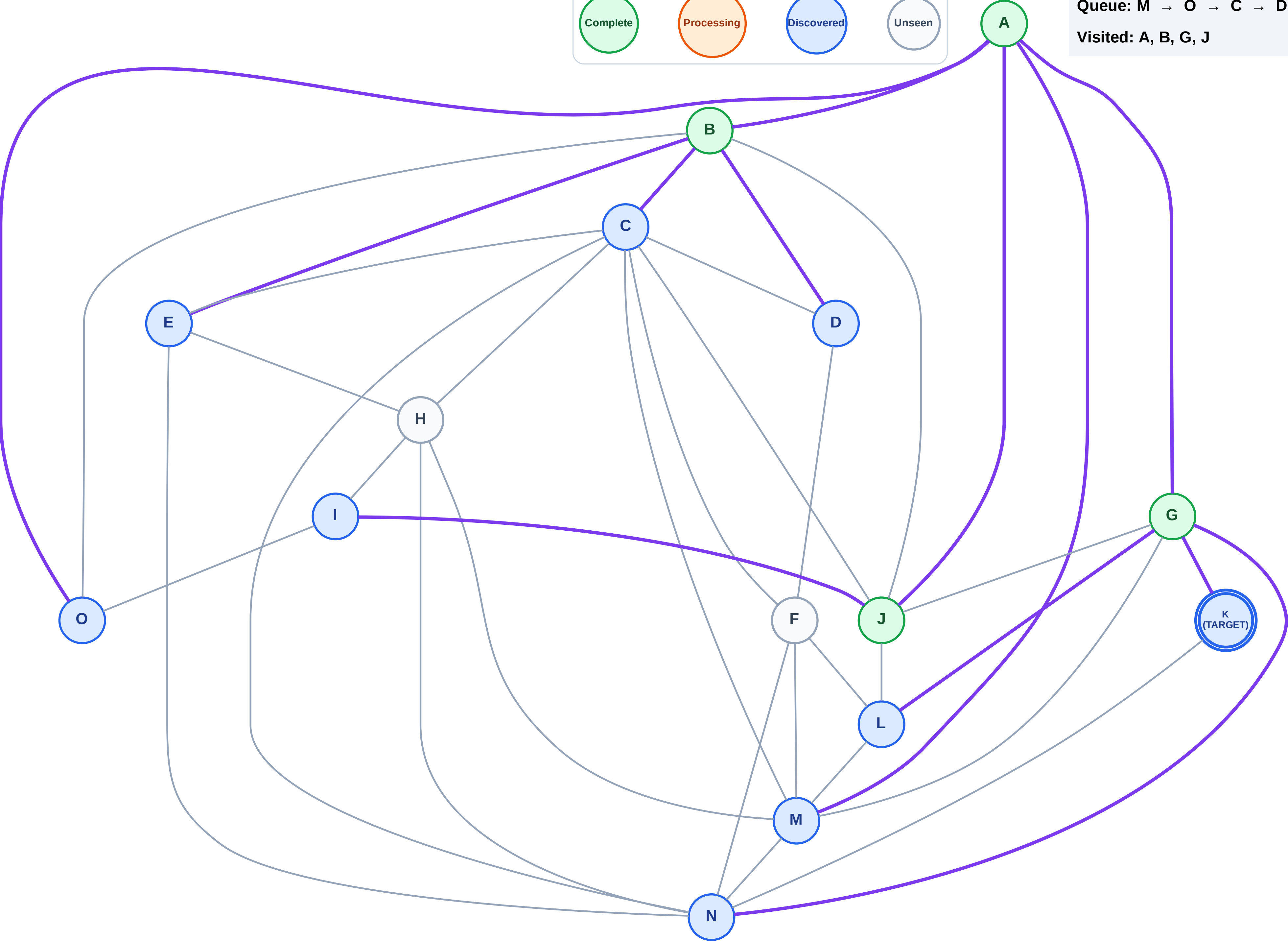
Processing

Discovered

Unseen

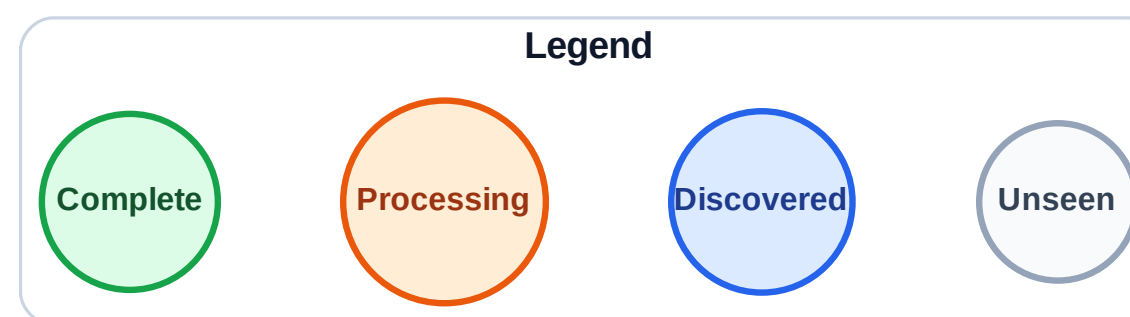
Queue: M → O → C → D → E → K → L → N → I

Visited: A, B, G, J



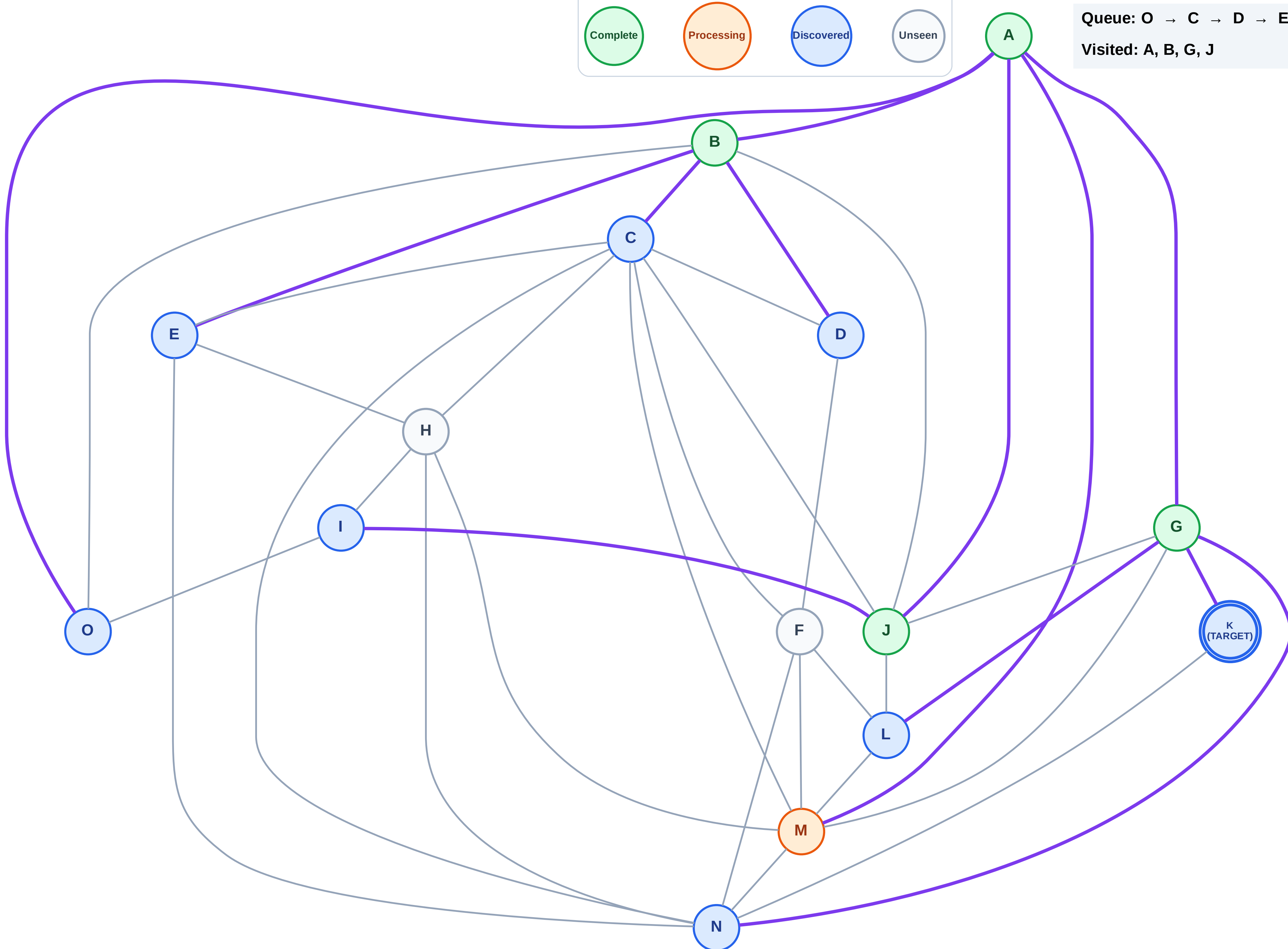
BFS Traversal

Step 10: Process node M



Queue: O → C → D → E → K → L → N → I

Visited: A, B, G, J



Step 11: Discover F , H from M

Step 11: Discover F , H from M

Complete

Processing

Discovered

Unseen

Visited: A, B, G, J, M

BFS Traversal

Step 12: Process node O

Legend

Complete

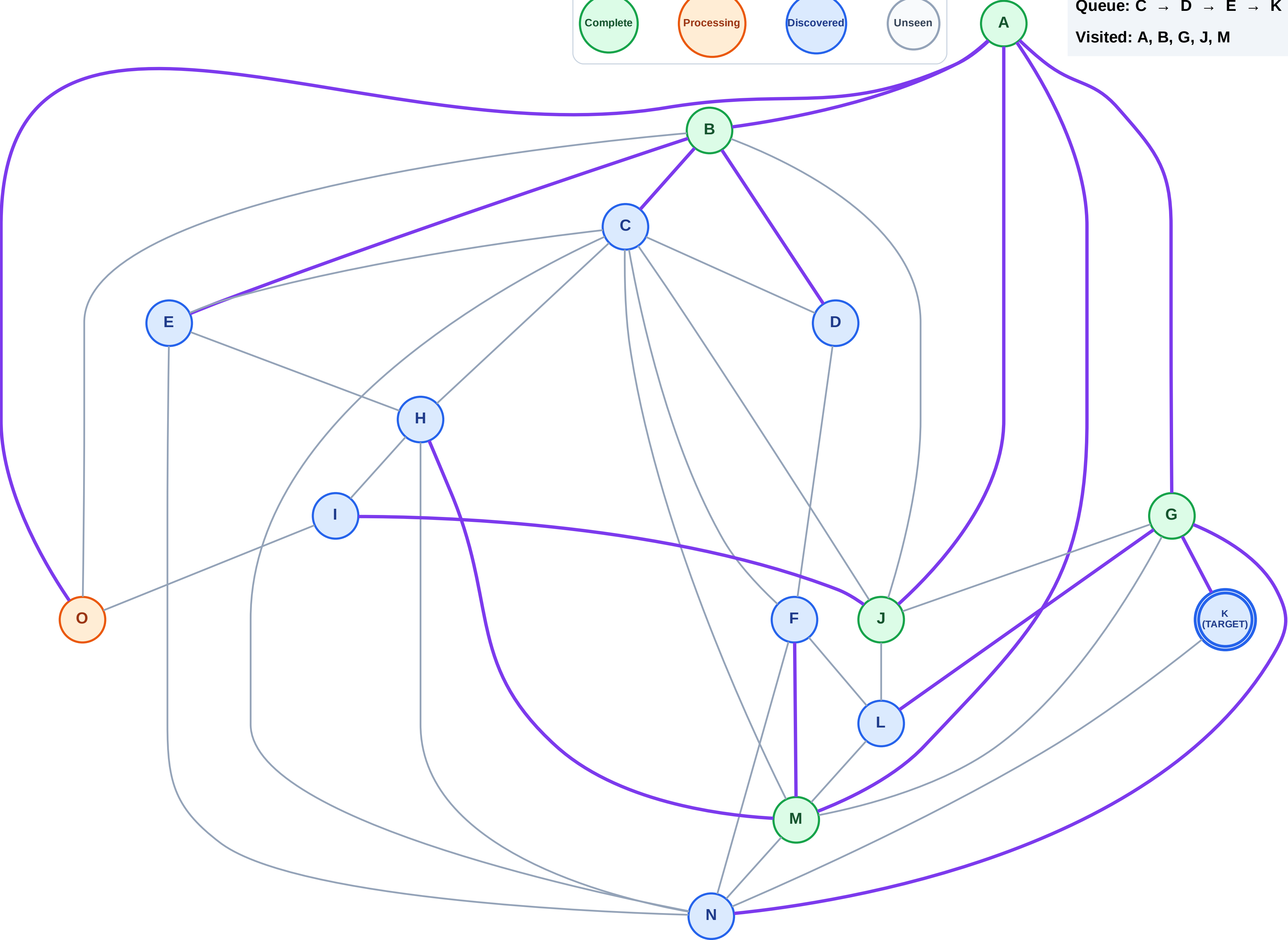
Processing

Discovered

Unseen

Queue: C → D → E → K → L → N → I → F → H

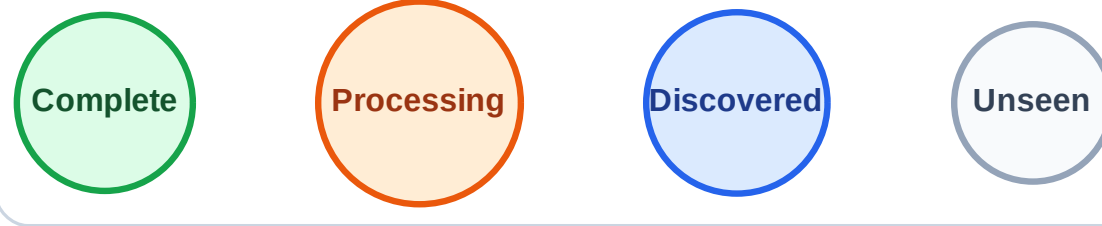
Visited: A, B, G, J, M



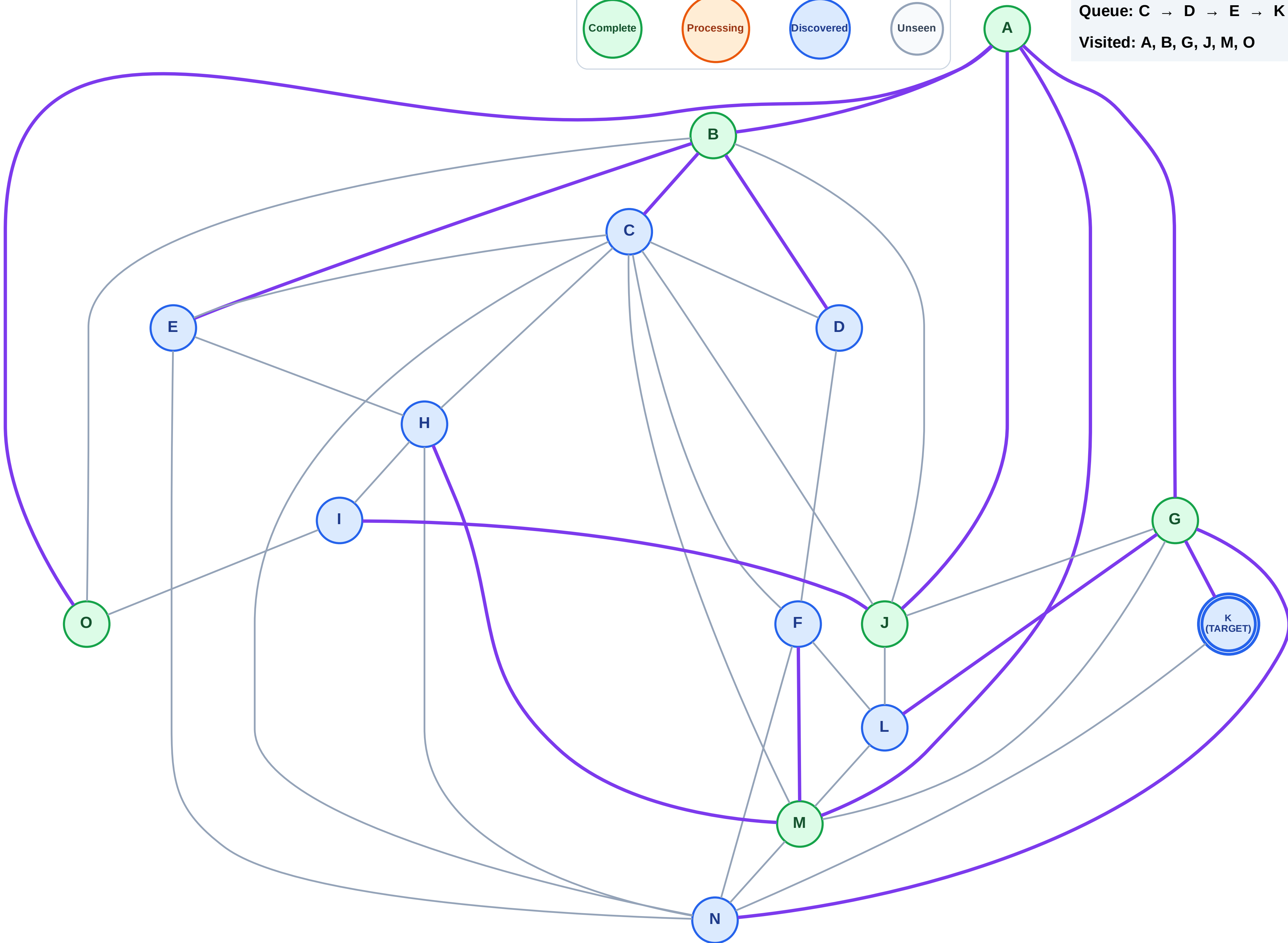
BFS Traversal

Step 13: No new neighbors from O

Legend



Queue: C → D → E → K → L → N → I → F → H
Visited: A, B, G, J, M, O



BFS Traversal

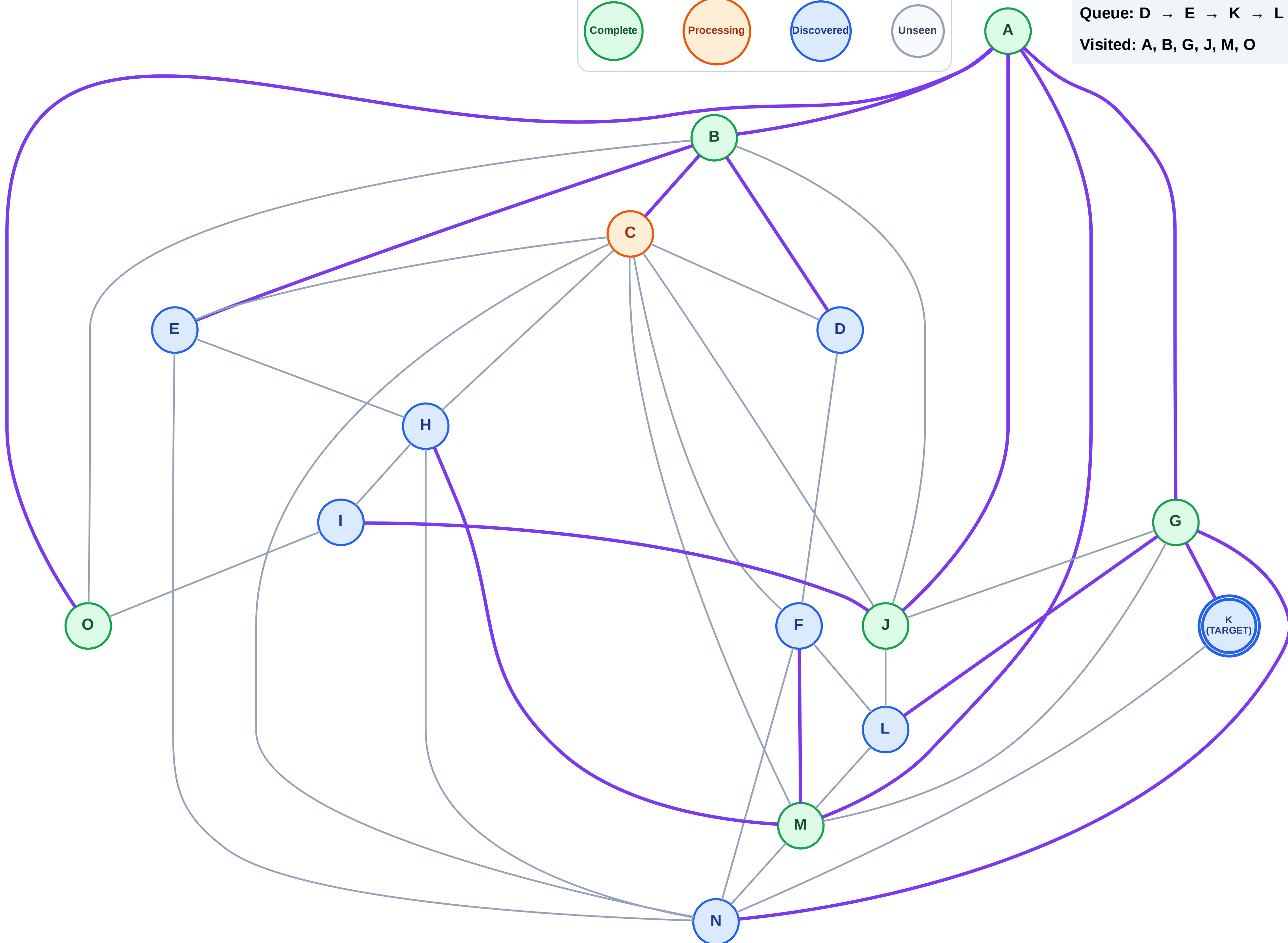
Step 14: Process node C

Legend



Queue: D → E → K → L → N → I → F → H

Visited: A, B, G, J, M, O



BFS Traversal

Step 15: No new neighbors from C

Legend



Complete



Processing

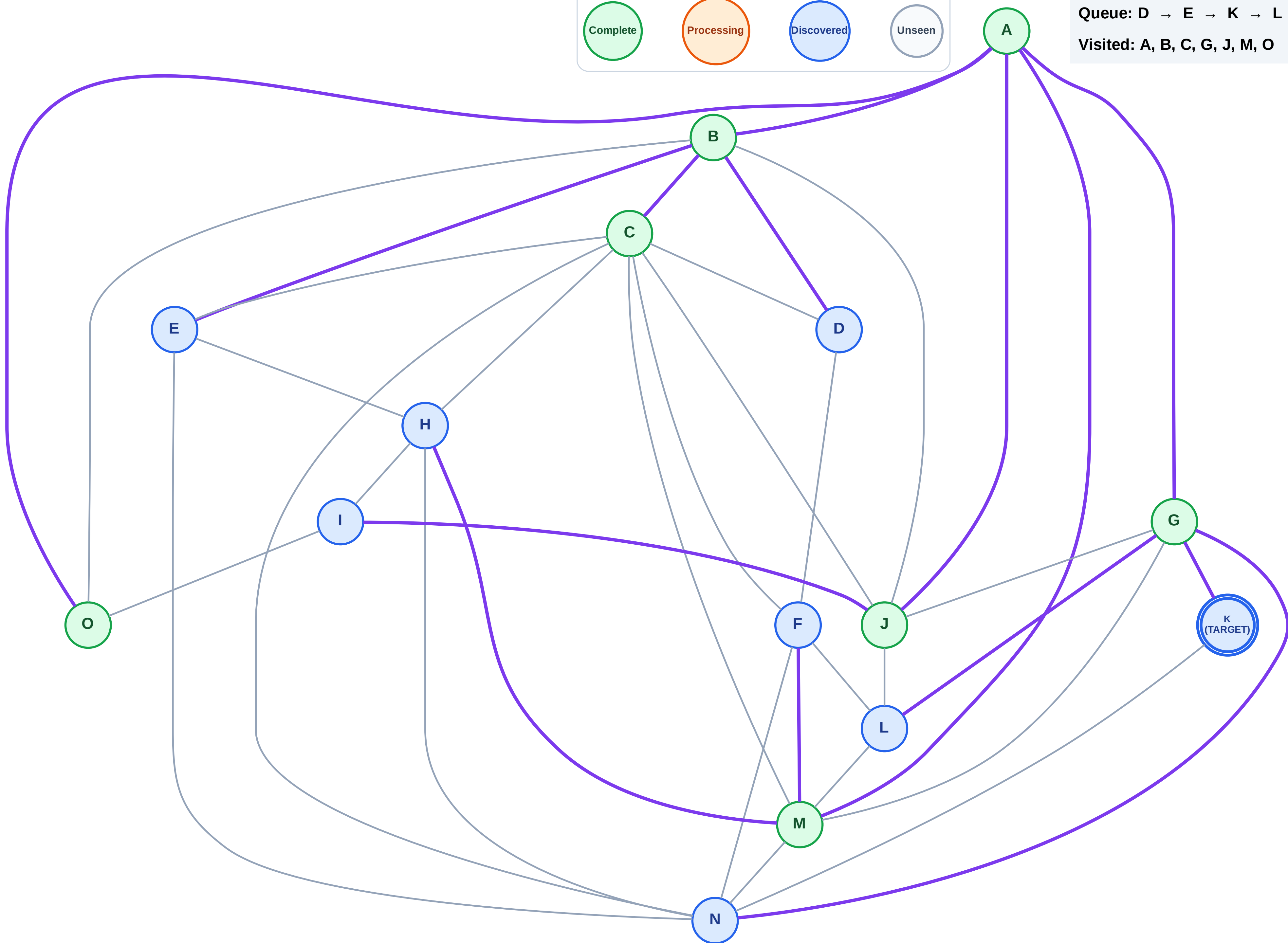


Discovered



Unseen

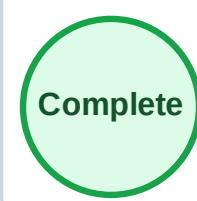
Queue: D → E → K → L → N → I → F → H
Visited: A, B, C, G, J, M, O



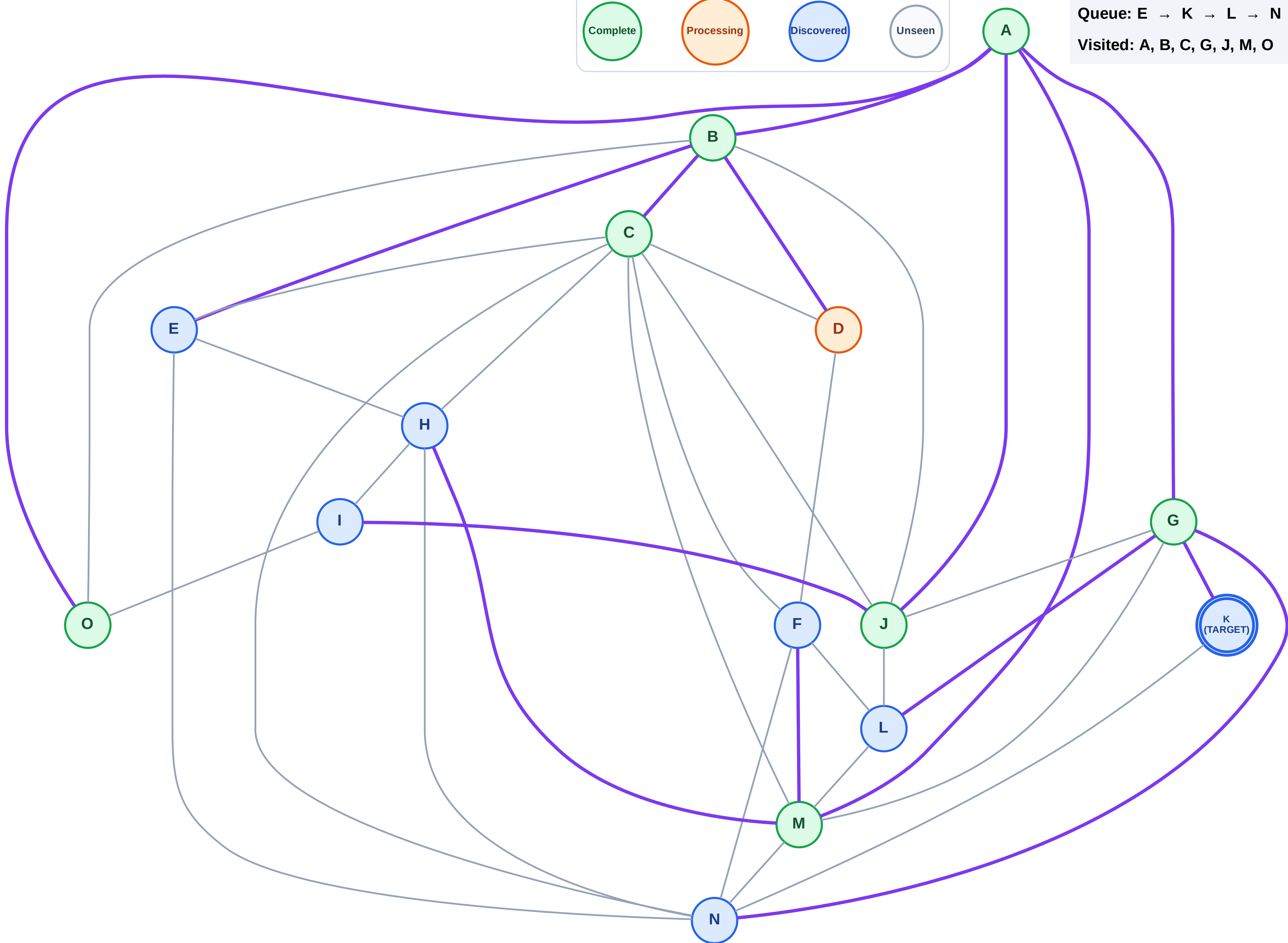
BFS Traversal

Step 16: Process node D

Legend



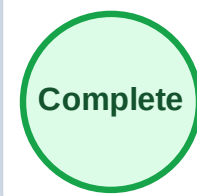
Queue: E → K → L → N → I → F → H
Visited: A, B, C, G, J, M, O



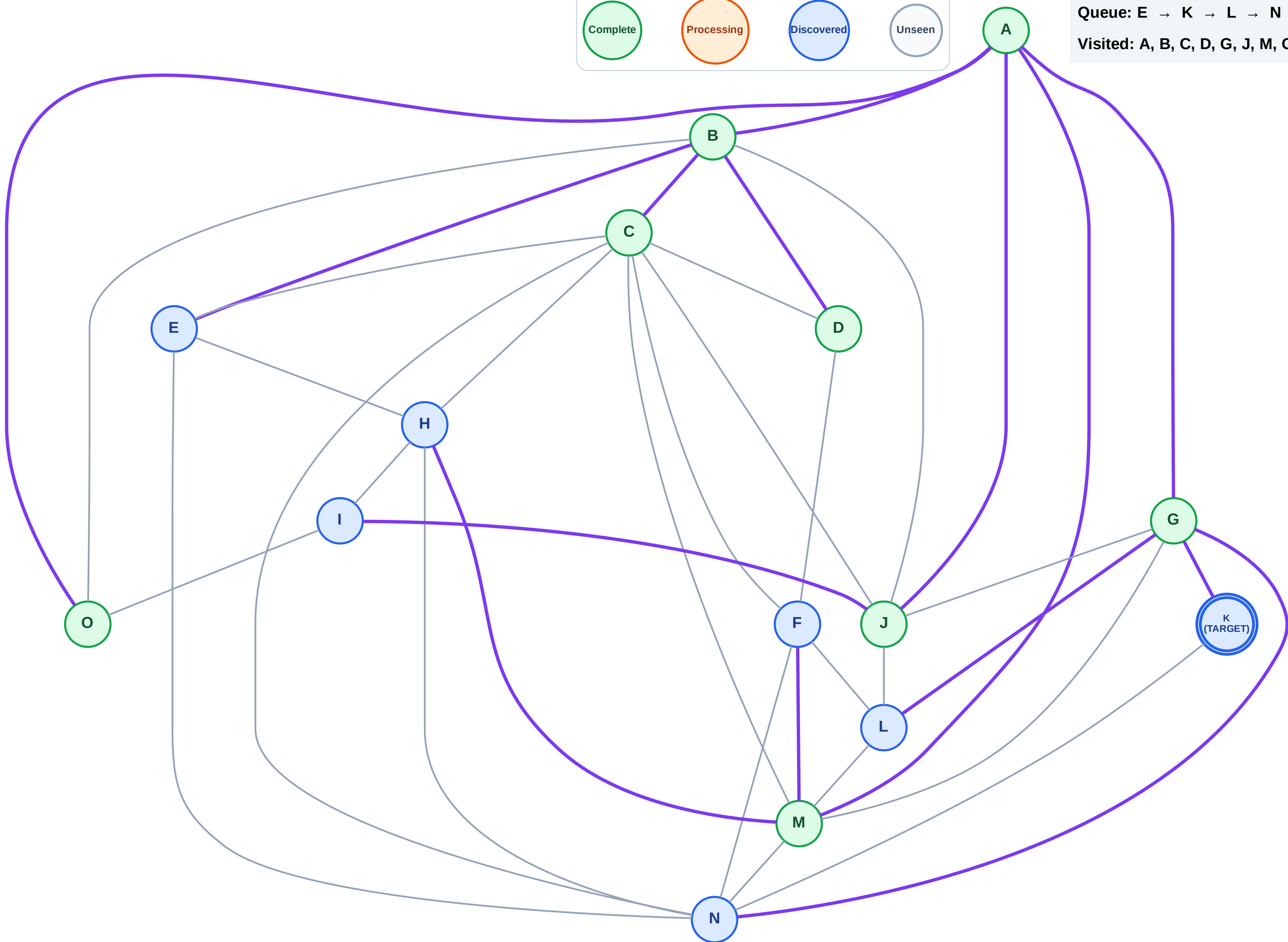
BFS Traversal

Step 17: No new neighbors from D

Legend



Queue: E → K → L → N → I → F → H
Visited: A, B, C, D, G, J, M, O



BFS Traversal

Step 18: Process node E

Legend



Complete



Processing



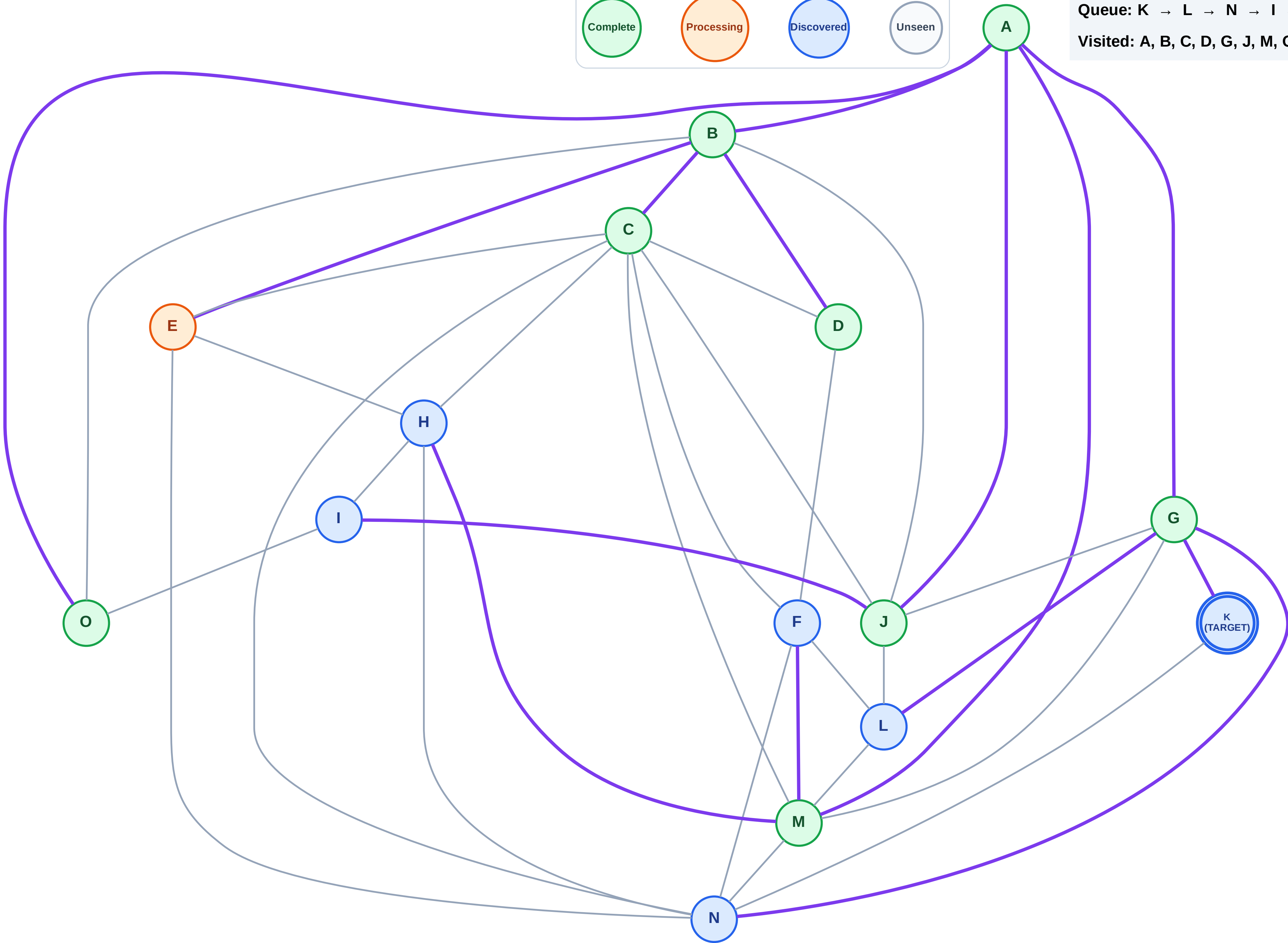
Discovered



Unseen

Queue: K → L → N → I → F → H

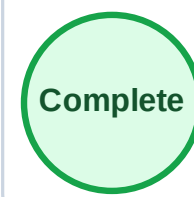
Visited: A, B, C, D, G, J, M, O



BFS Traversal

Step 19: No new neighbors from E

Legend



Complete



Processing



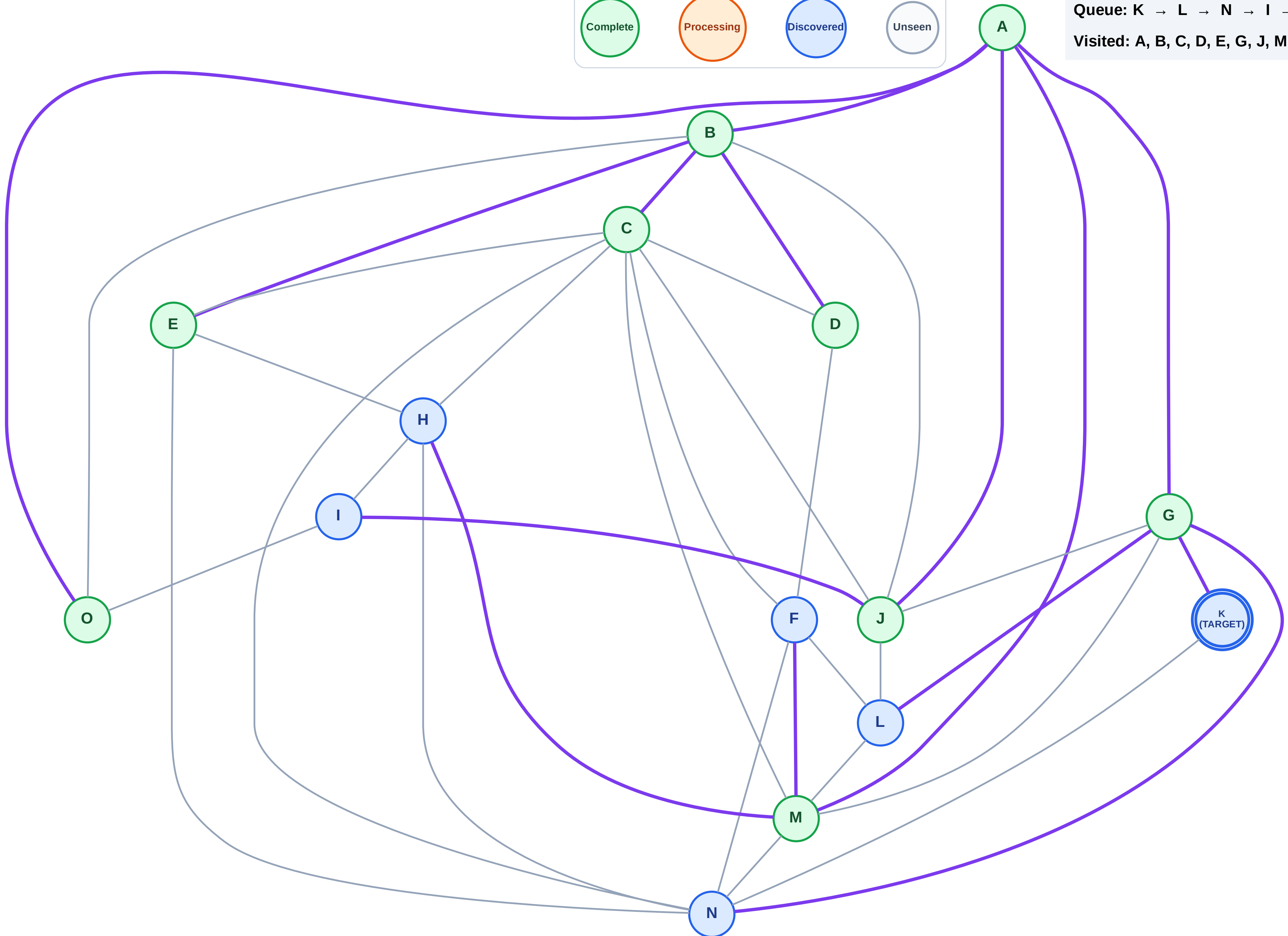
Discovered



Unseen

Queue: K → L → N → I → F → H

Visited: A, B, C, D, E, G, J, M, O



BFS Traversal

Step 20: Process node K

Legend



Complete



Processing



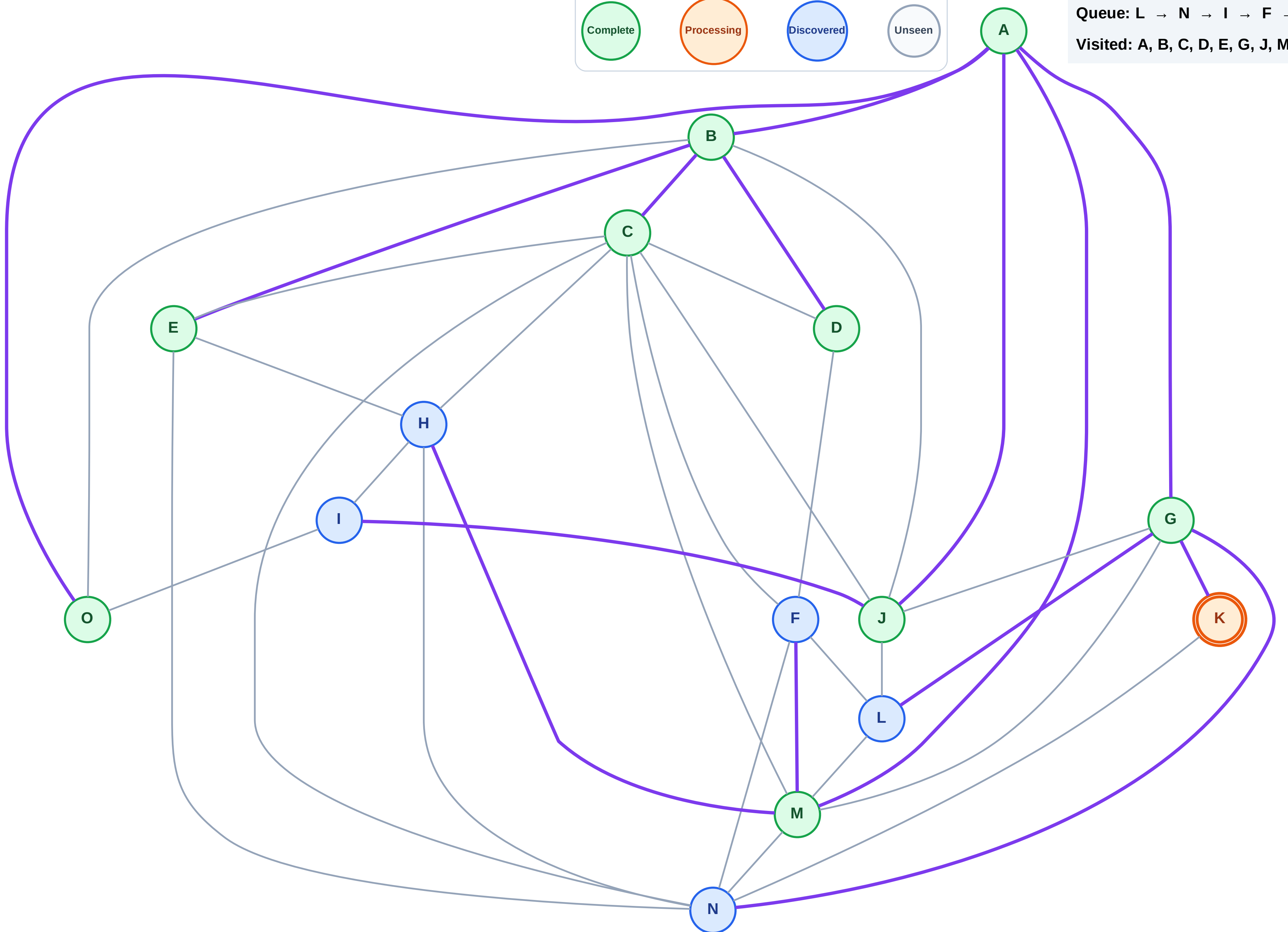
Discovered



Unseen

Queue: L → N → I → F → H

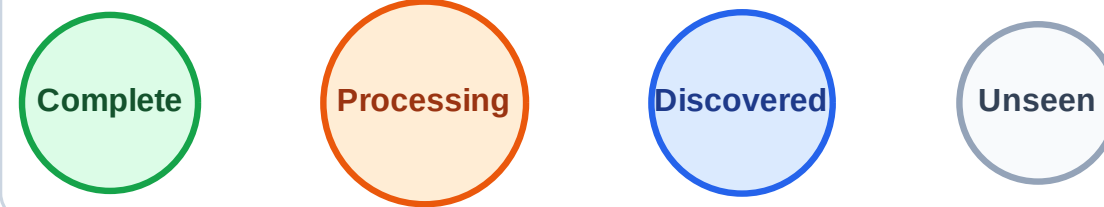
Visited: A, B, C, D, E, G, J, M, O



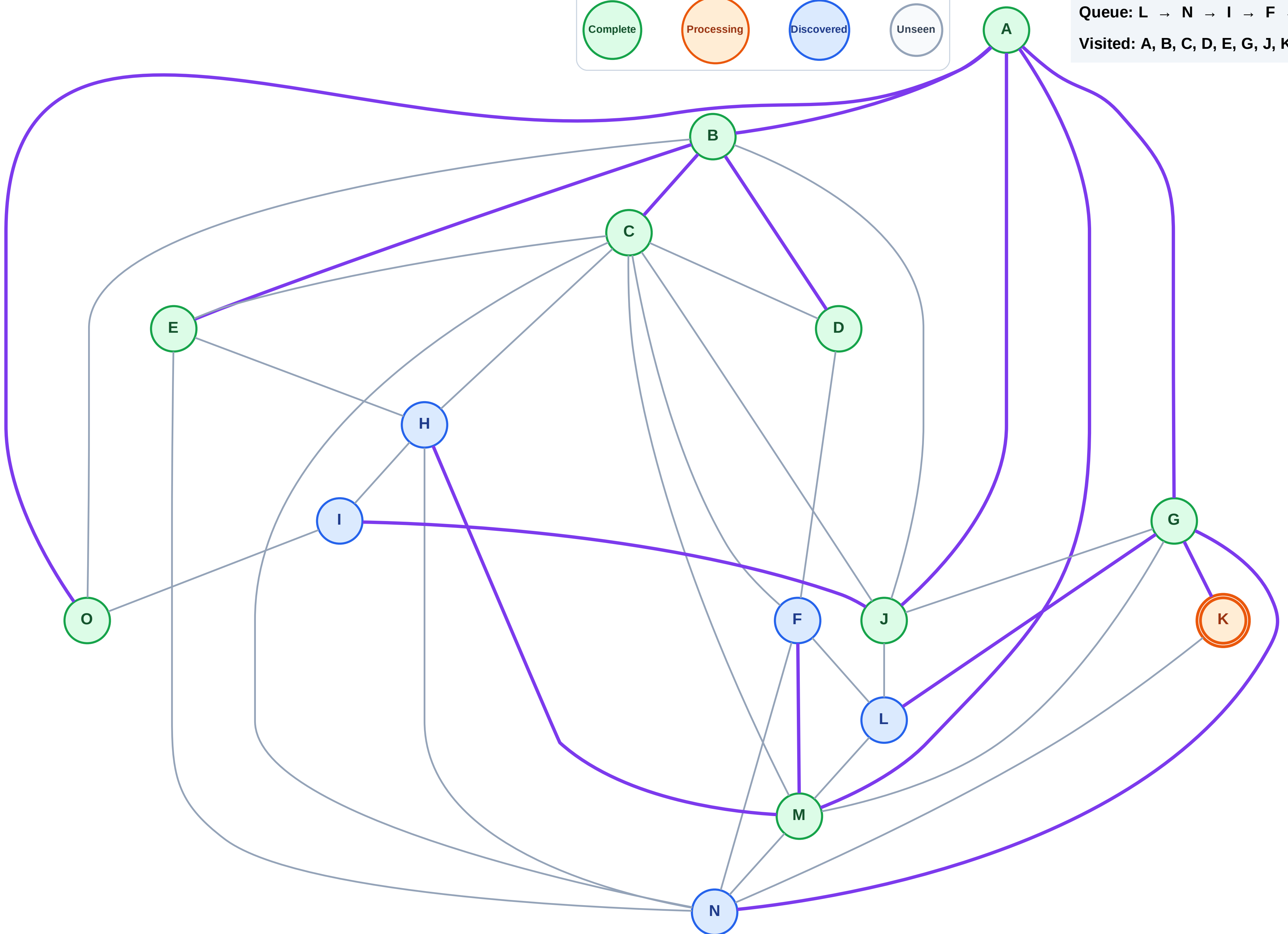
BFS Traversal

Step 21: TARGET NODE K REACHED! Path (2 hops): A → G → K

Legend



Queue: L → N → I → F → H
Visited: A, B, C, D, E, G, J, K, M, O



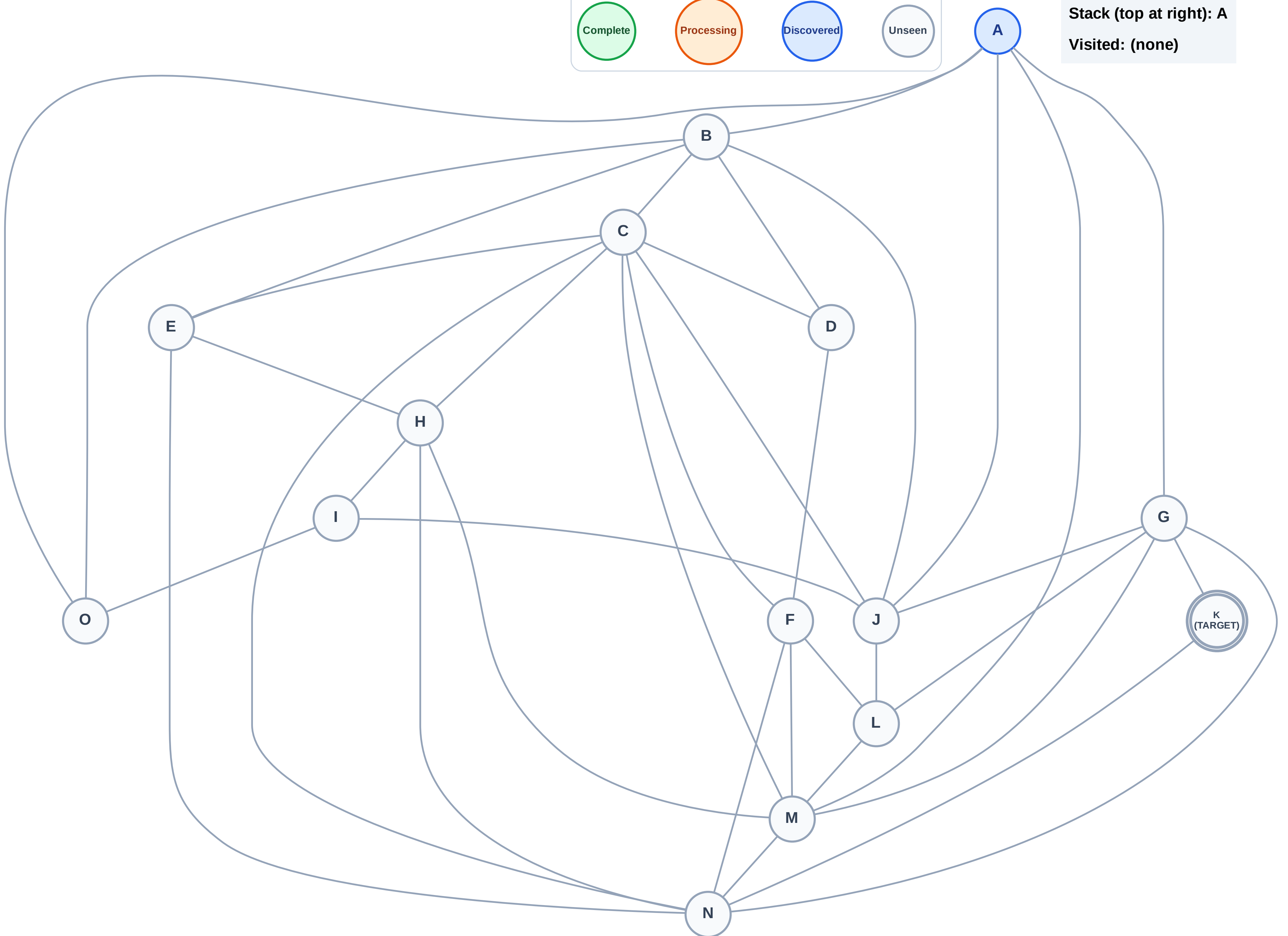
DFS Traversal

Step 1: Start at A (Seeking target node K)

Legend



Stack (top at right): A
Visited: (none)



DFS Traversal

Step 2: Process node A

Legend

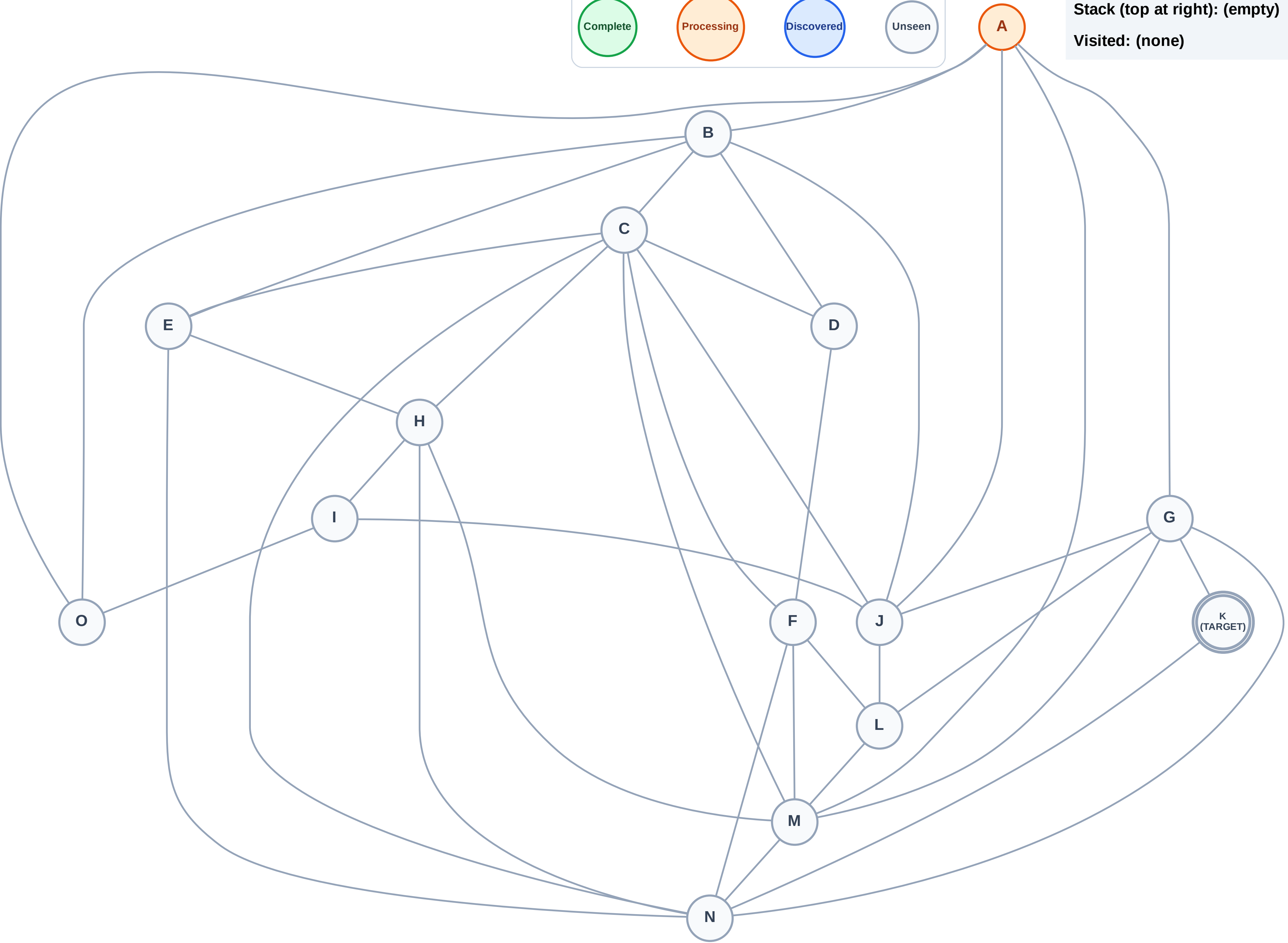
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

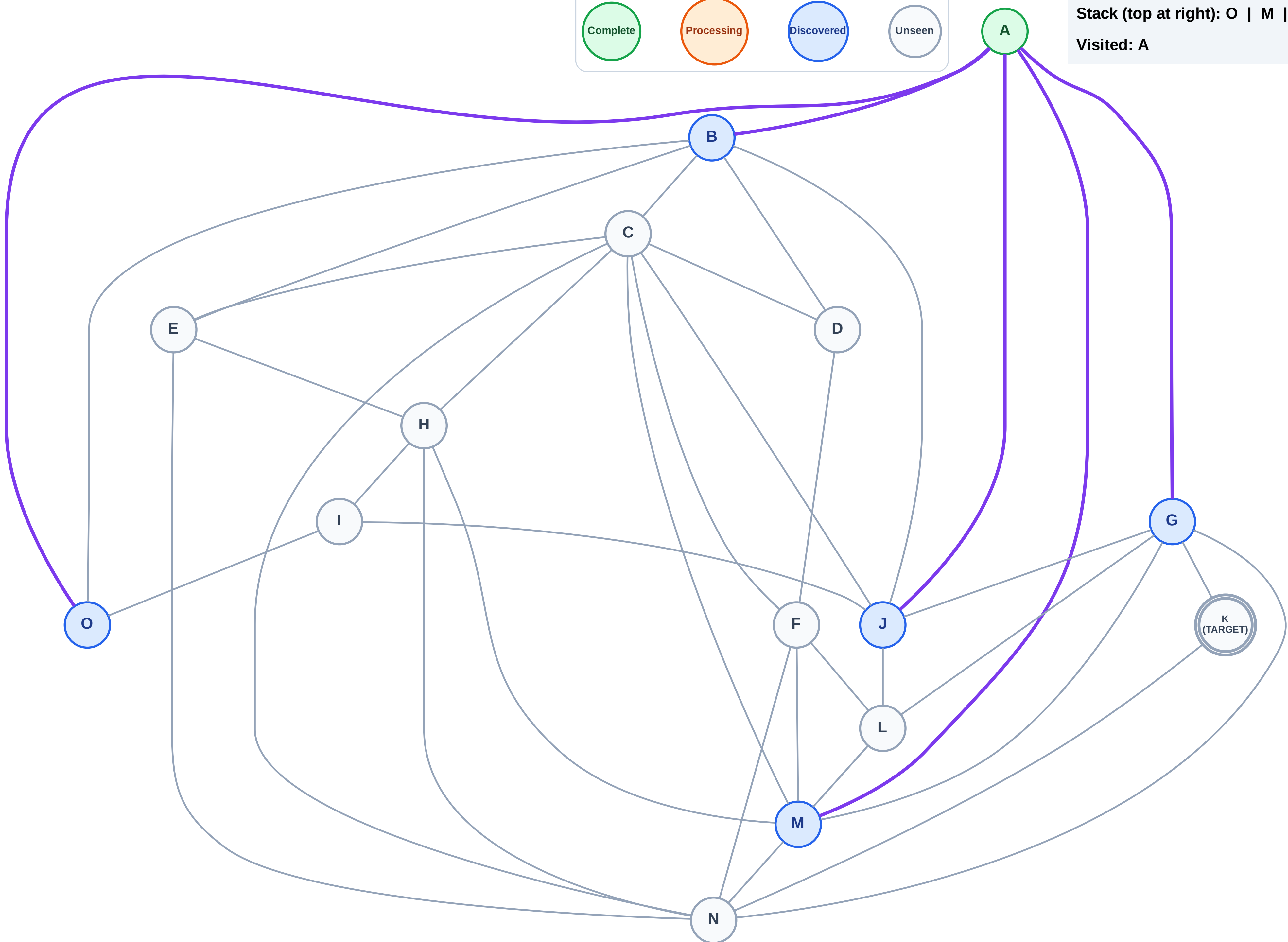
Step 3: Discover O, M, J, G, B from A

Legend



Stack (top at right): O | M | J | G | B

Visited: A



DFS Traversal

Step 4: Process node B

Legend



Complete



Processing



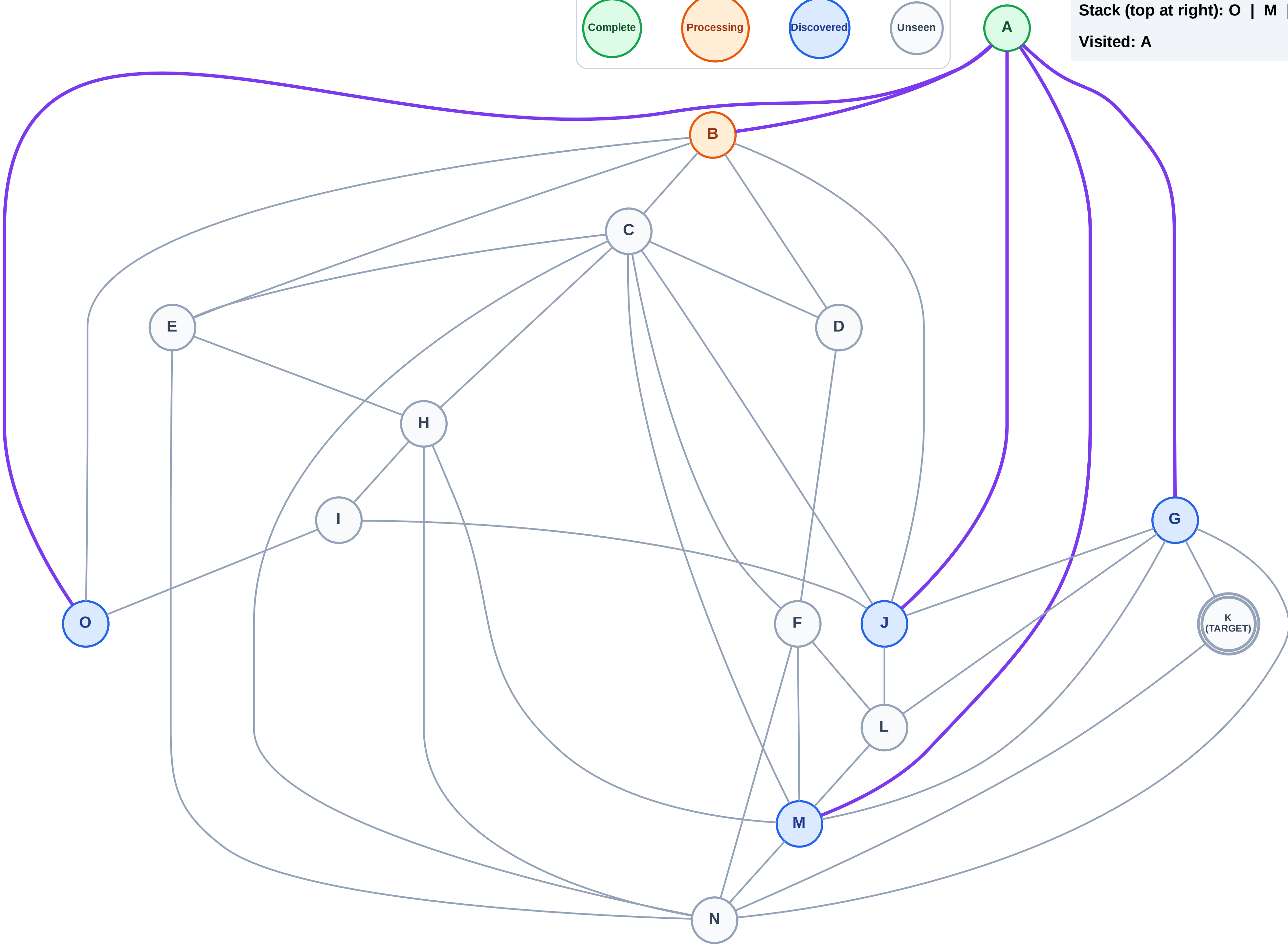
Discovered



Unseen

Stack (top at right): O | M | J | G

Visited: A



DFS Traversal

Step 5: Discover E, D, C from B

Legend



Complete



Processing



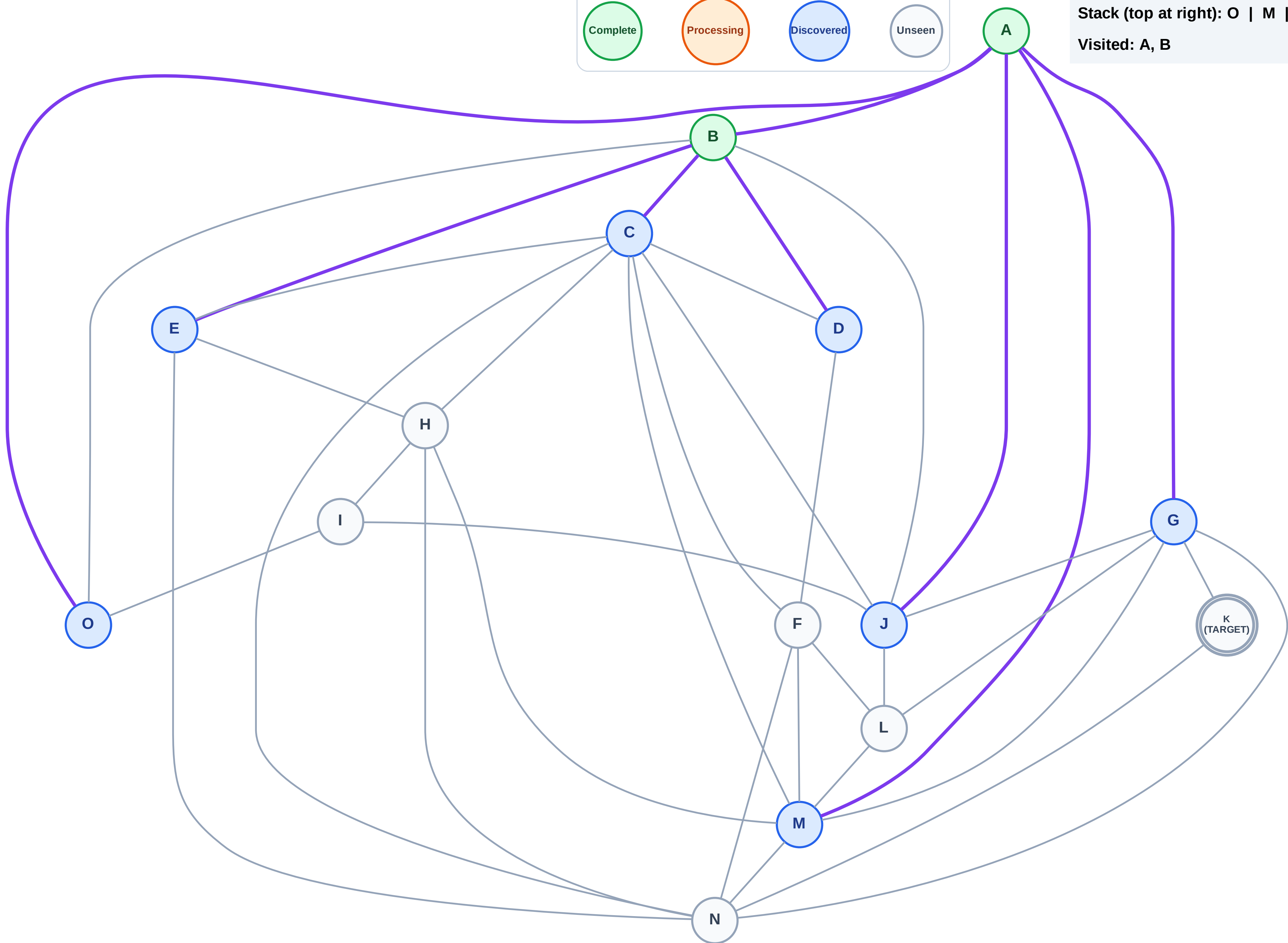
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C

Legend

Complete

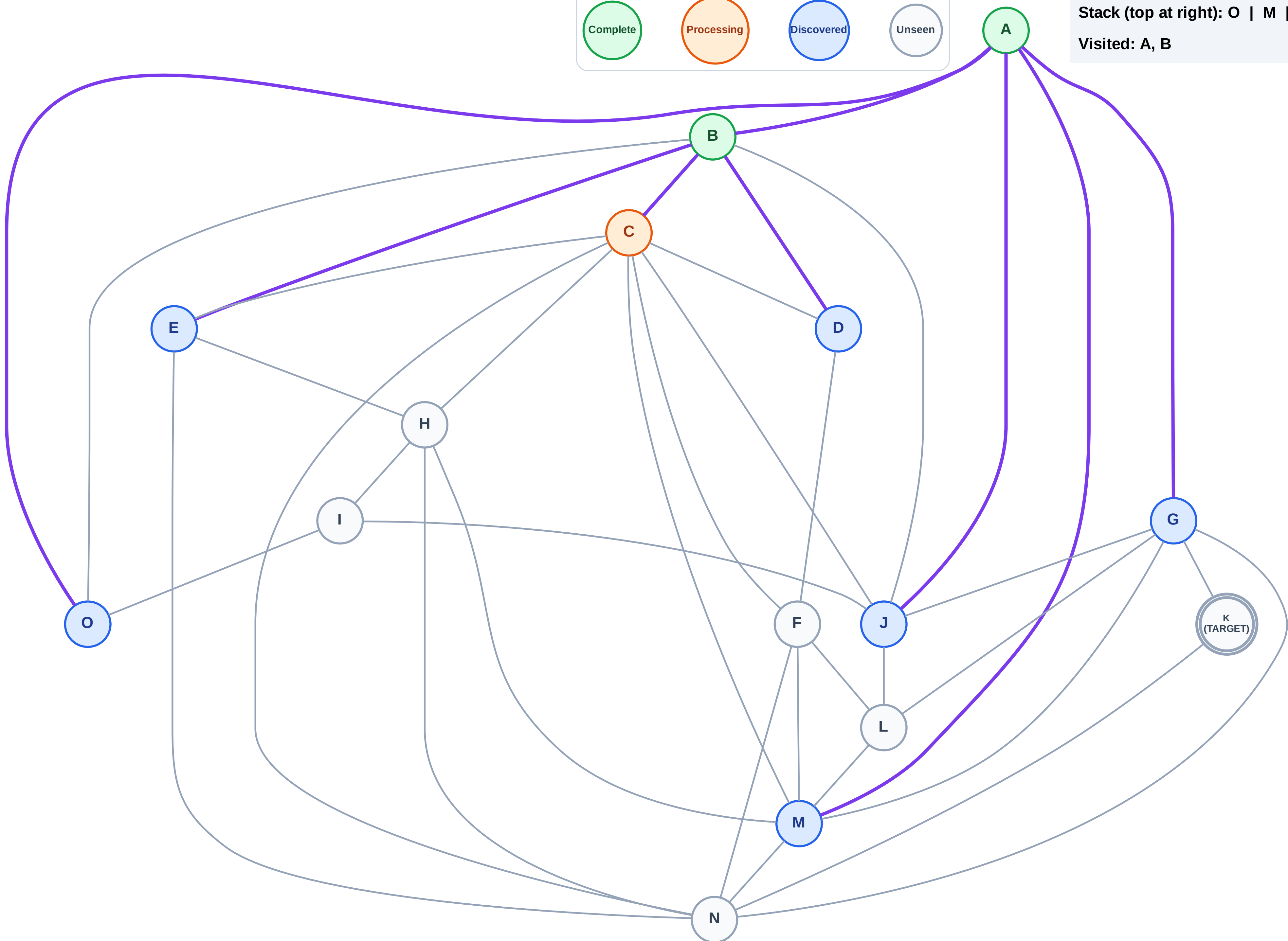
Processing

Discovered

Unseen

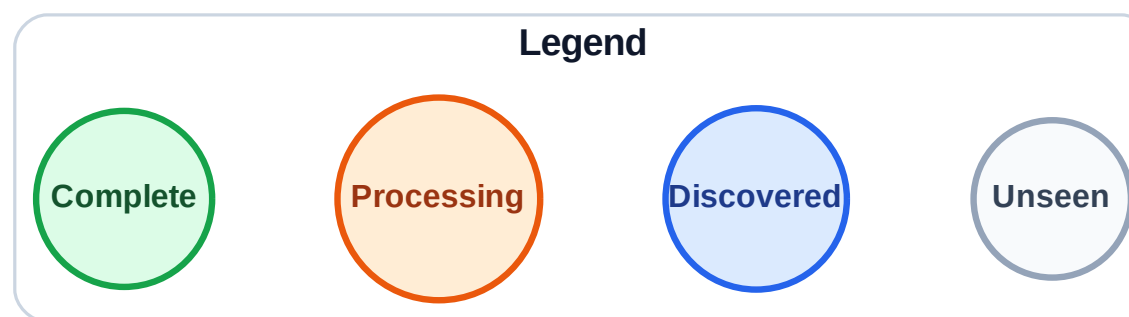
Stack (top at right): O | M | J | G | E | D

Visited: A, B



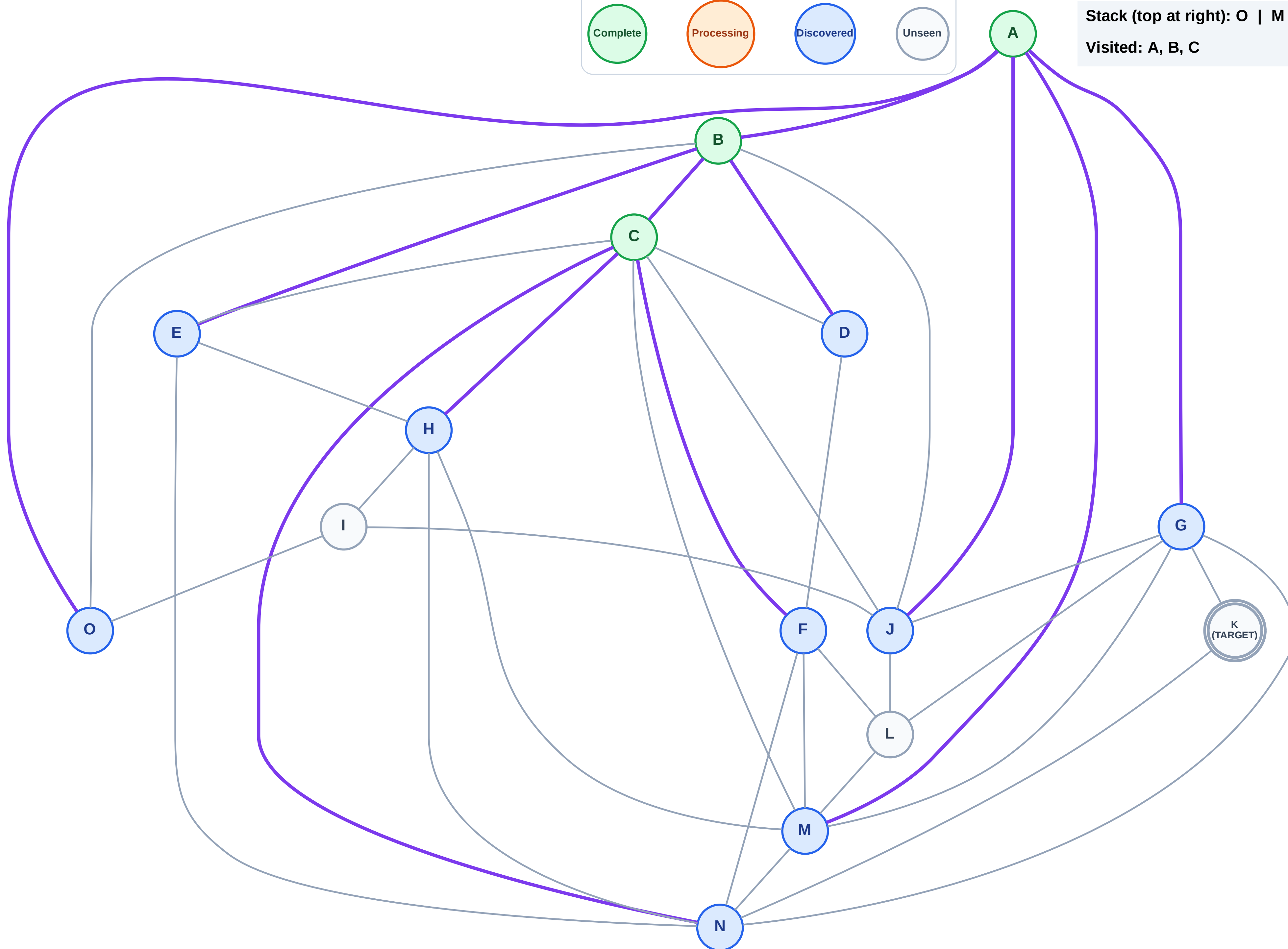
DFS Traversal

Step 7: Discover N, H, F from C



Stack (top at right): O | M | J | G | E | D | N | H | F

Visited: A, B, C



Step 8: Process node F

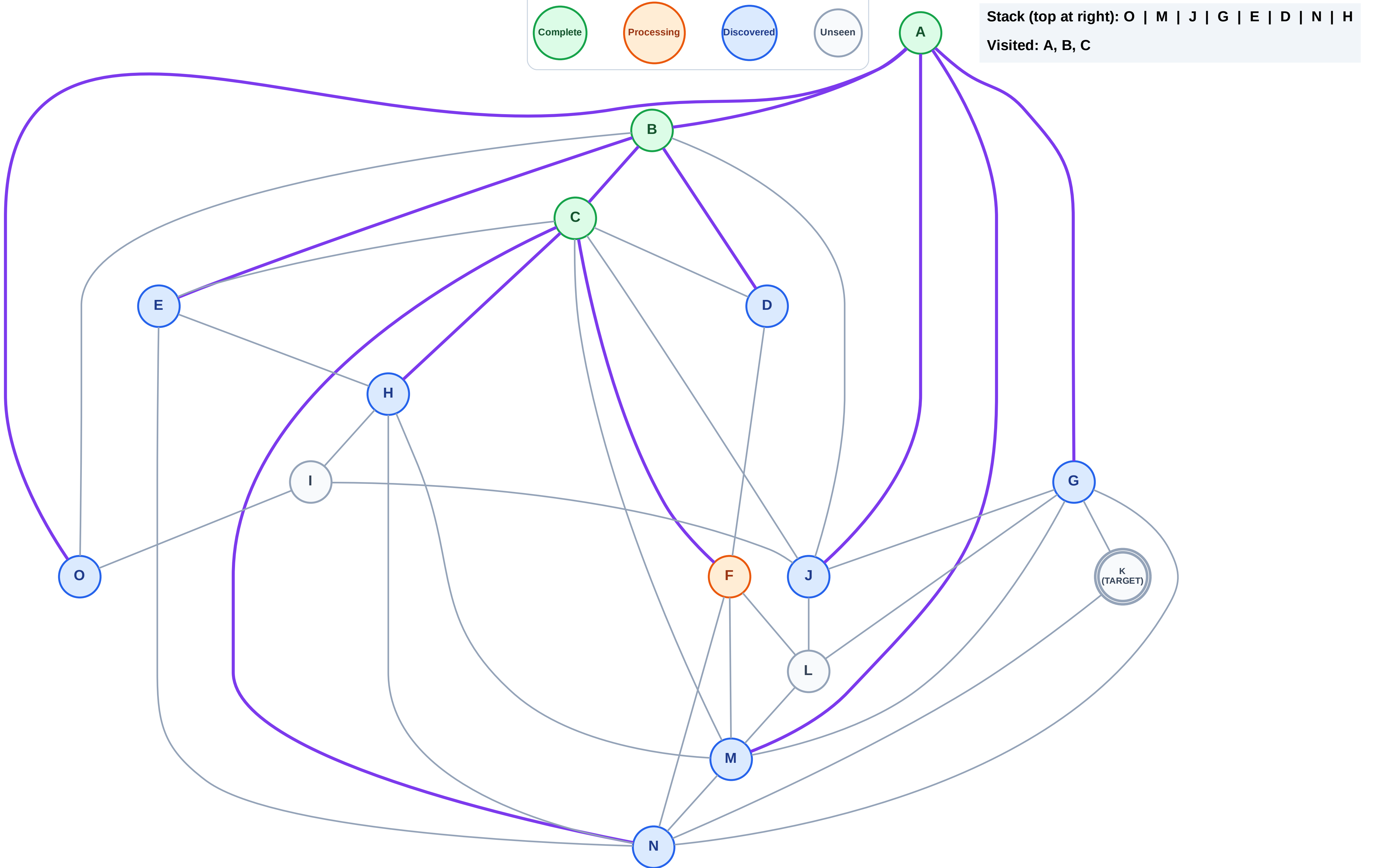
Step 8: Process node F

The diagram illustrates the four states of a node in a graph search algorithm:

- Complete**: Represented by a green circle.
- Processing**: Represented by an orange circle.
- Discovered**: Represented by a blue circle.
- Unseen**: Represented by a grey circle.

Unseen

Visited: A, B, C



DFS Traversal

Step 9: Discover L from F

Legend

Complete

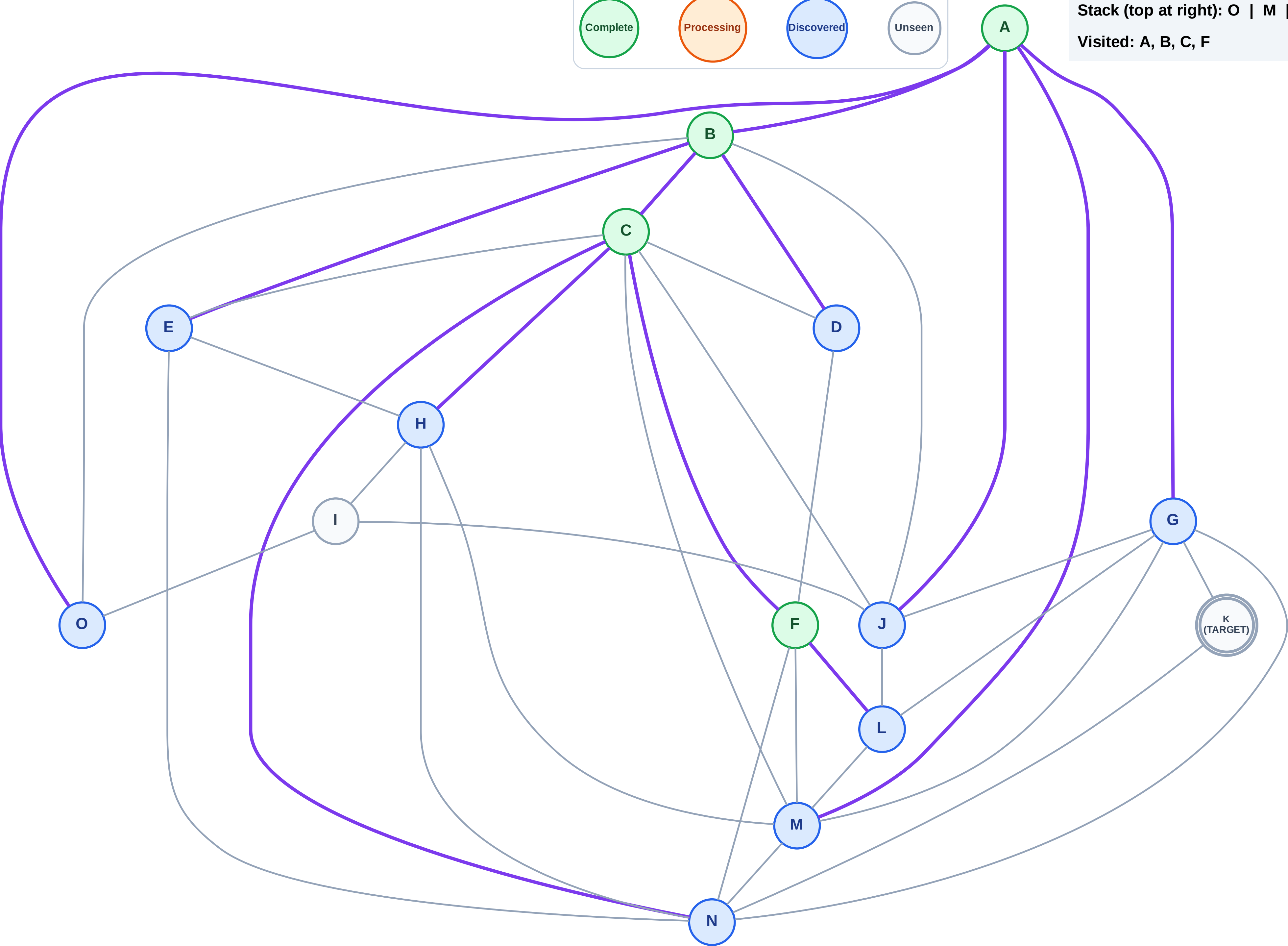
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N | H | L

Visited: A, B, C, F



DFS Traversal

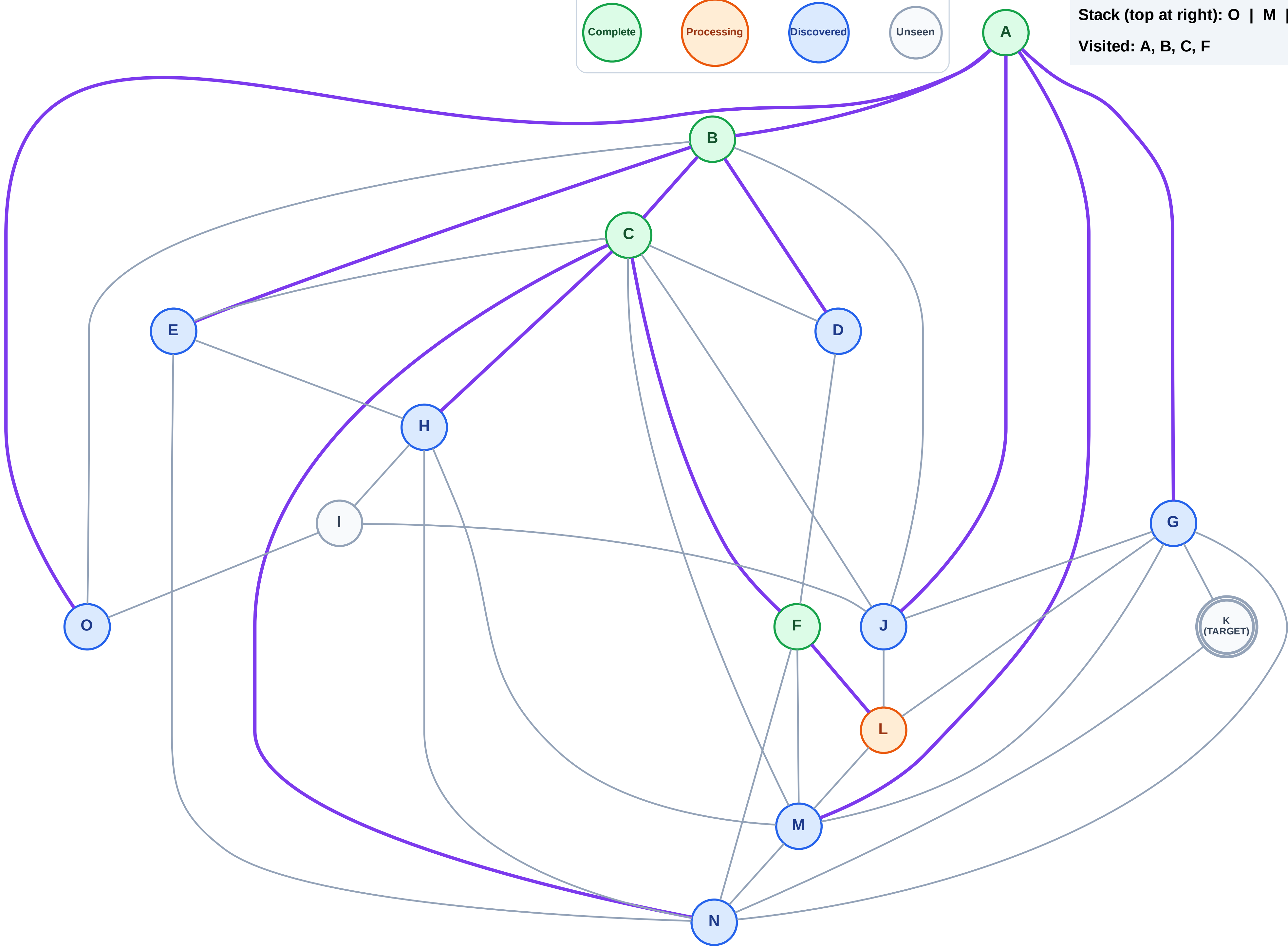
Step 10: Process node L

Legend



Stack (top at right): O | M | J | G | E | D | N | H

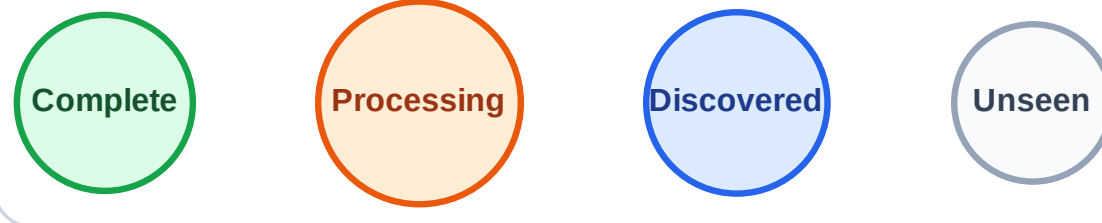
Visited: A, B, C, F



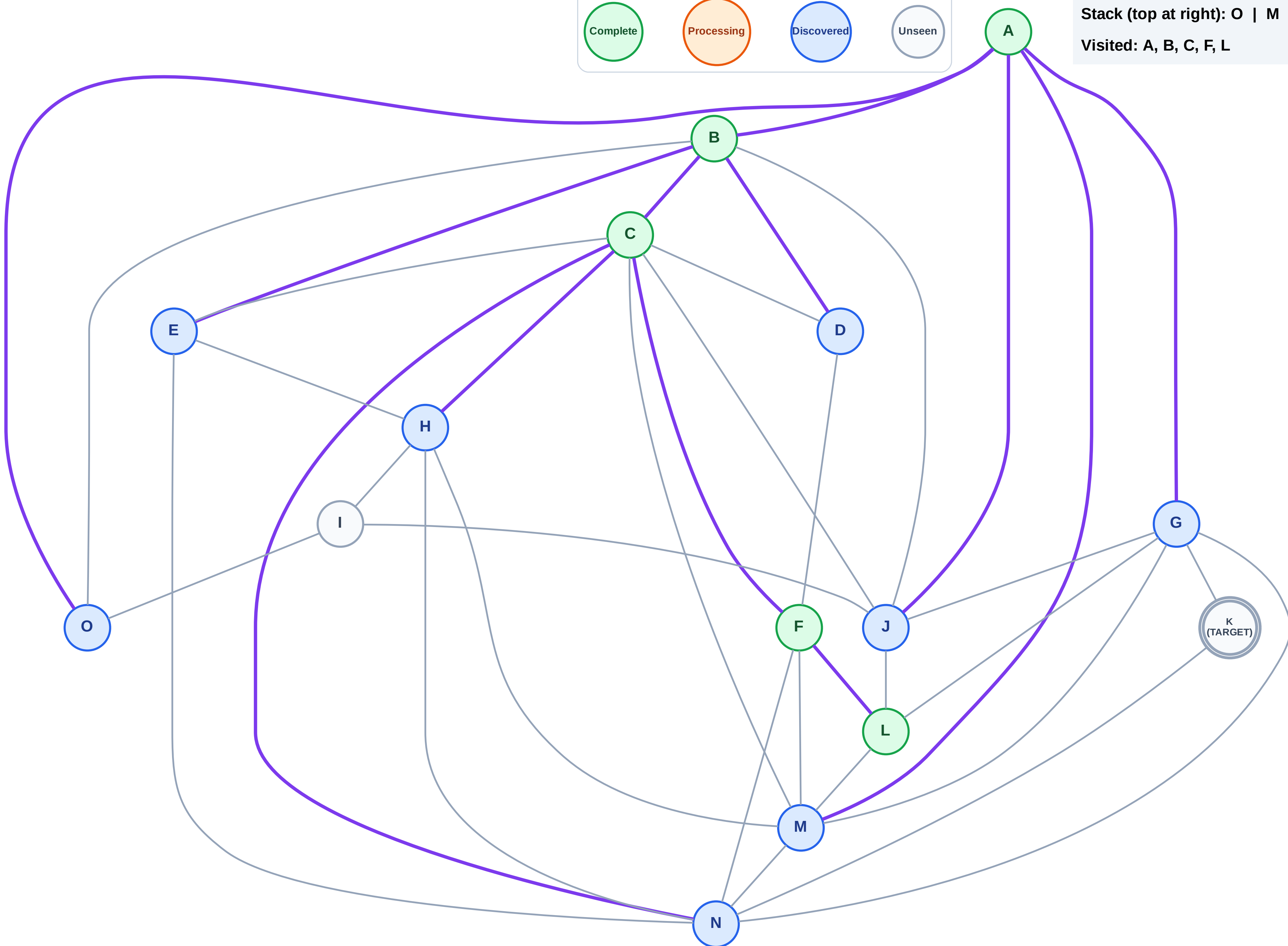
DFS Traversal

Step 11: No new neighbors from L

Legend



Stack (top at right): O | M | J | G | E | D | N | H
Visited: A, B, C, F, L



Step 12: Process node H

Legend



DFS Traversal

Step 13: Discover I from H

Legend

Complete

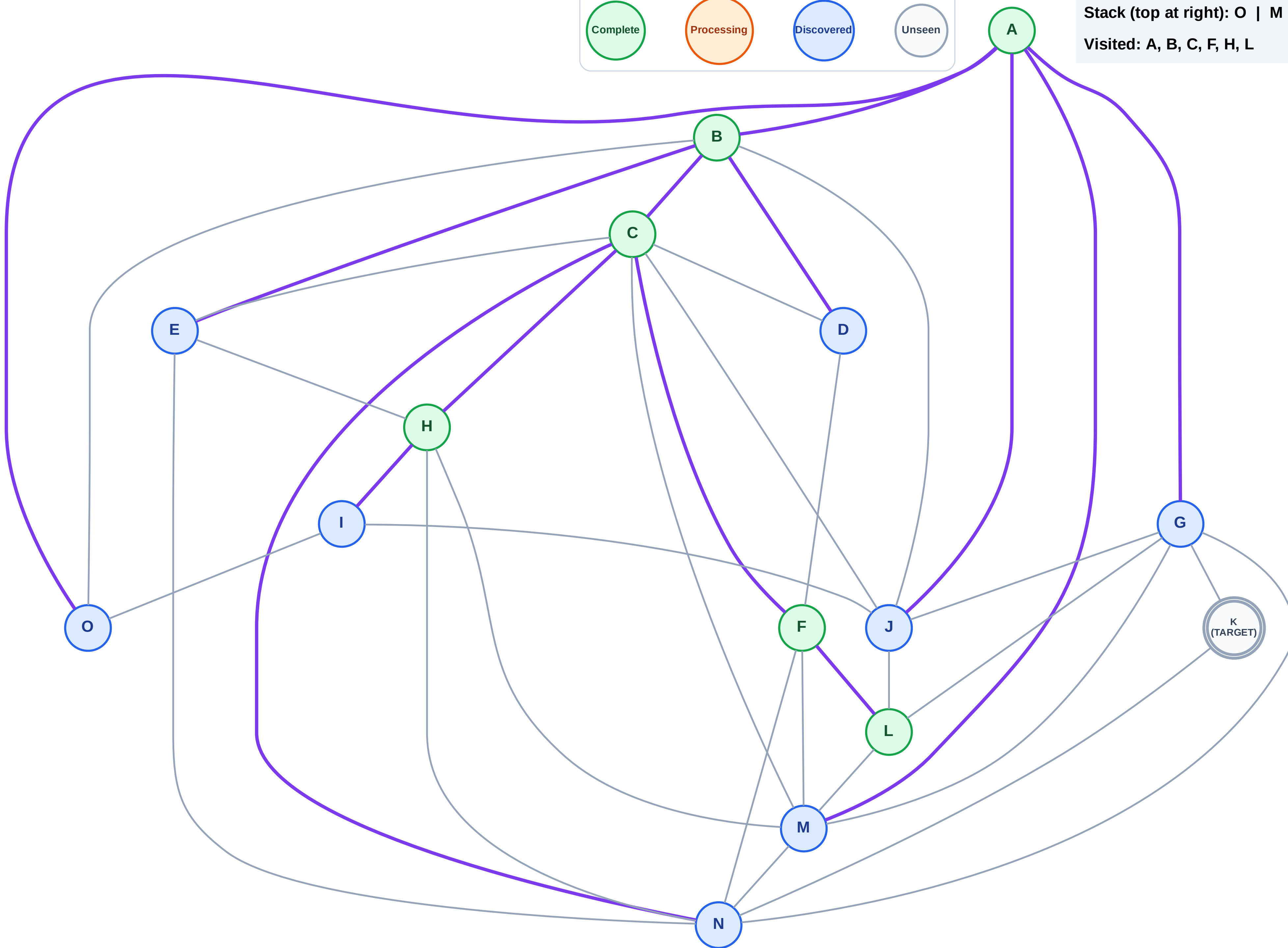
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D | N | I

Visited: A, B, C, F, H, L



DFS Traversal

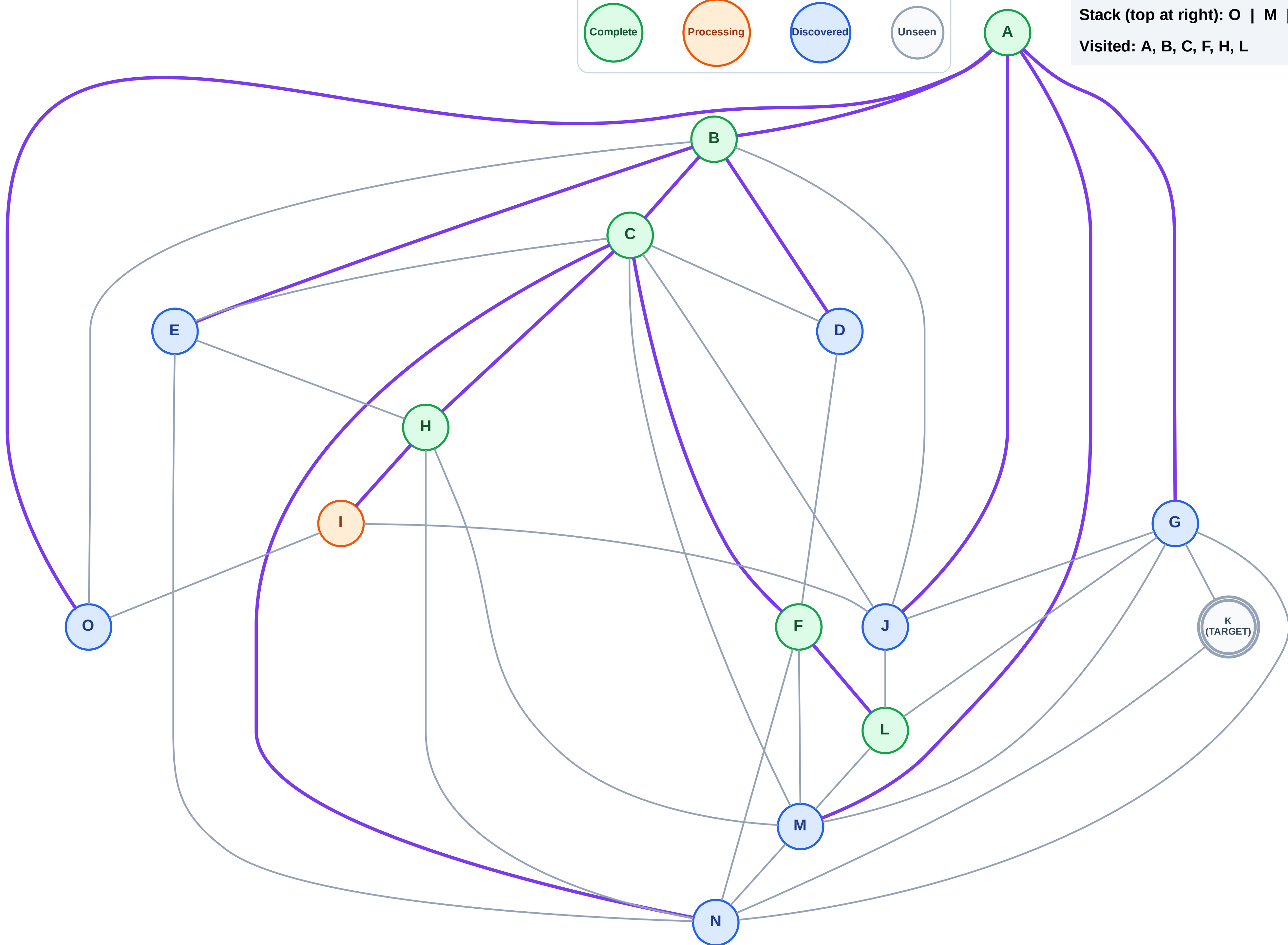
Step 14: Process node I

Legend



Stack (top at right): O | M | J | G | E | D | N

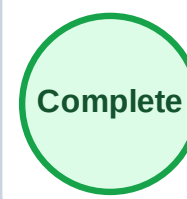
Visited: A, B, C, F, H, L



DFS Traversal

Step 15: No new neighbors from I

Legend



Complete



Processing



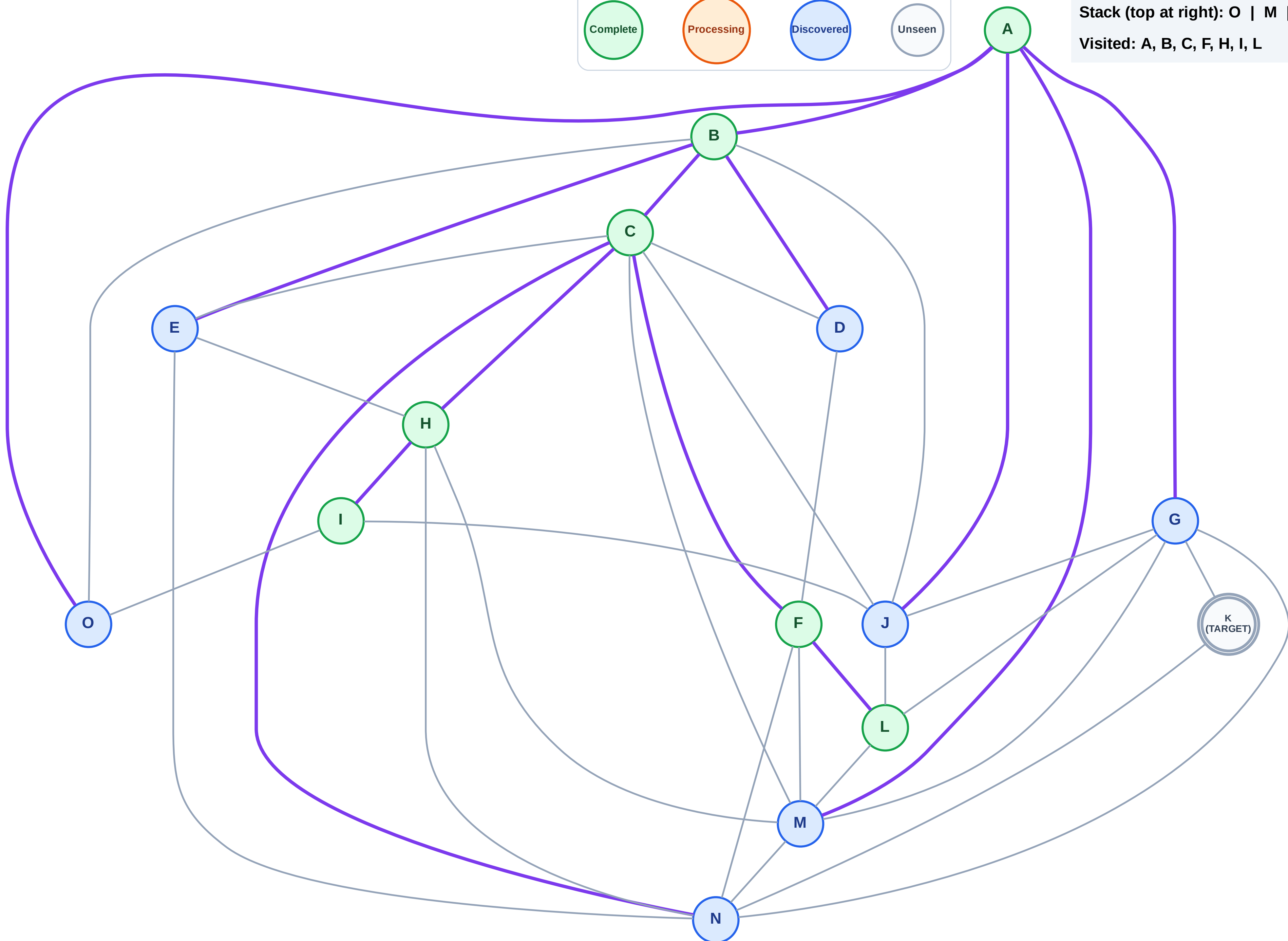
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | N

Visited: A, B, C, F, H, I, L



Step 16: Process node N

Step 16: Process node N

Legend

Complete

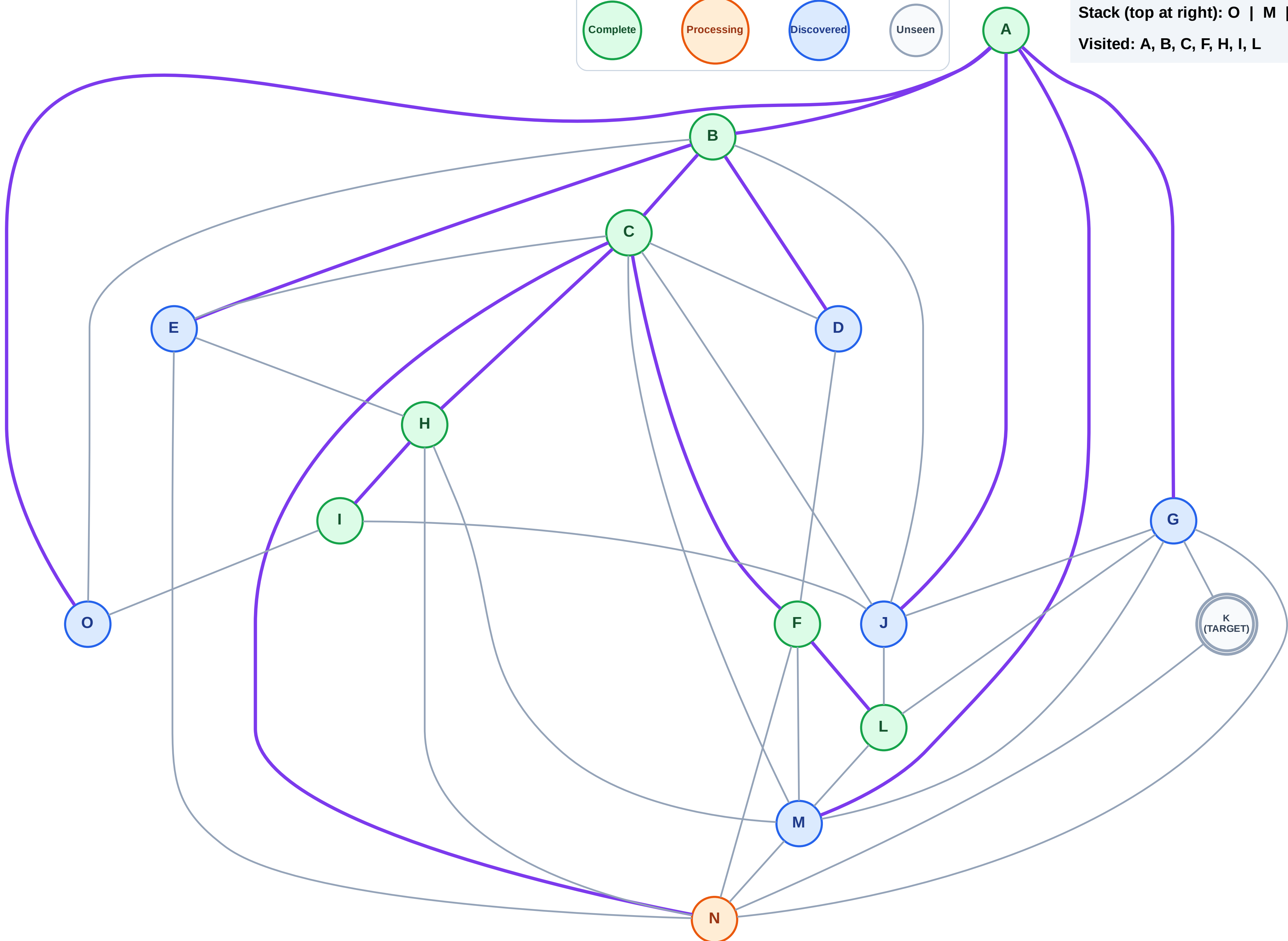
Processing

Discovered

Unseen

Stack (top at right): O | M | J | G | E | D

Visited: A, B, C, F, H, I, L



DFS Traversal

Step 17: Discover K from N

Legend



Complete



Processing



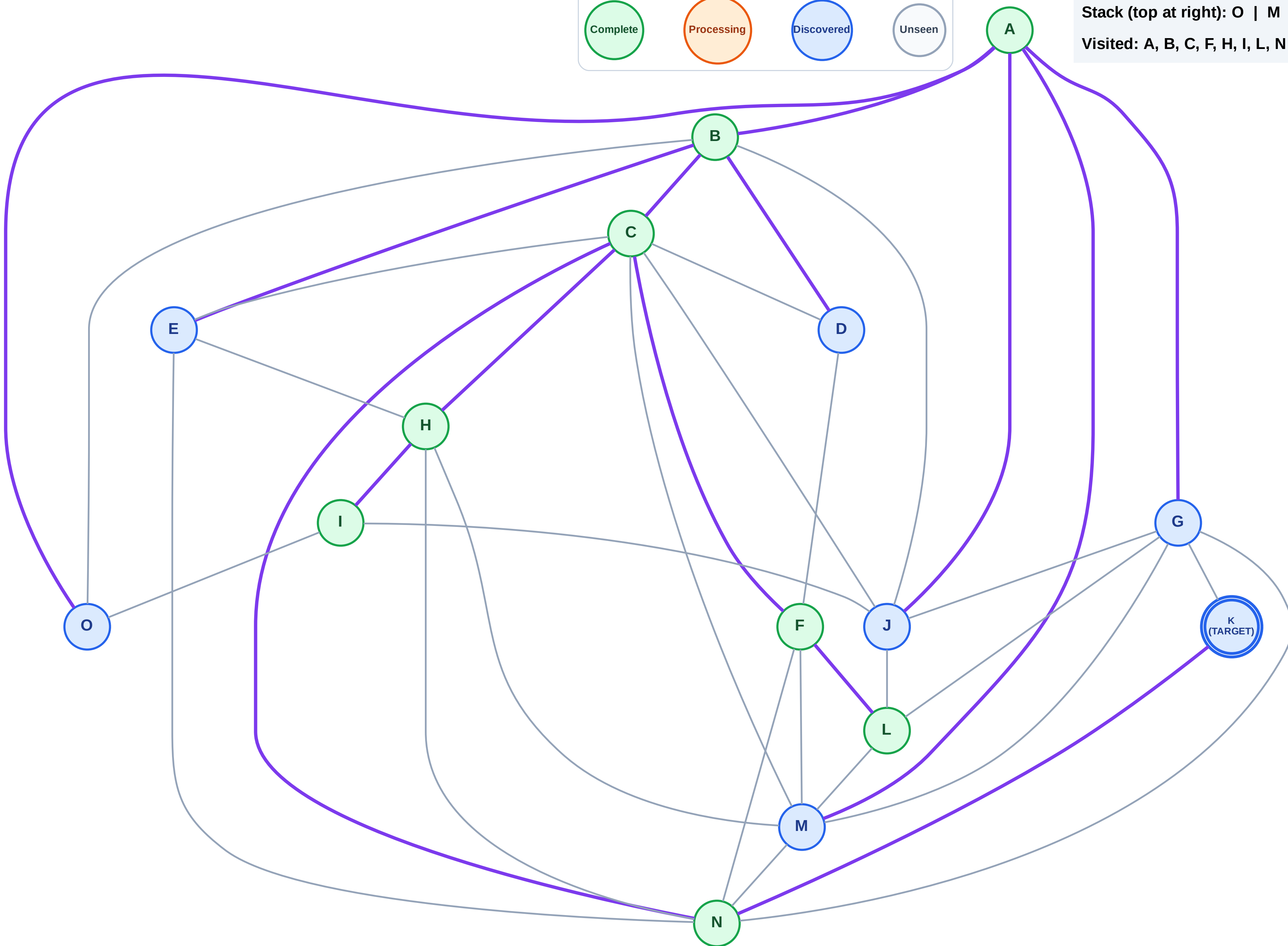
Discovered



Unseen

Stack (top at right): O | M | J | G | E | D | K

Visited: A, B, C, F, H, I, L, N



DFS Traversal

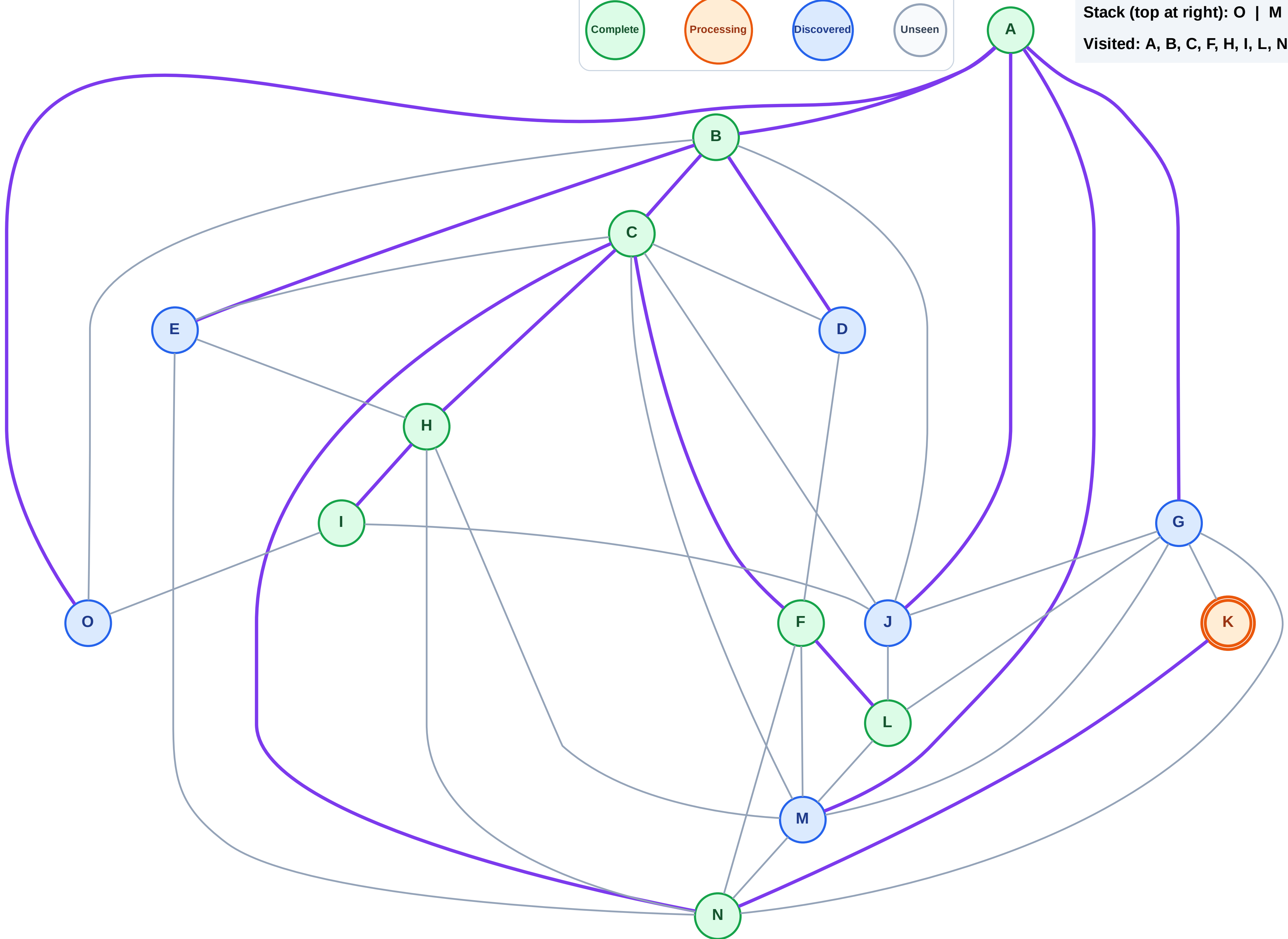
Step 18: Process node K

Legend



Stack (top at right): O | M | J | G | E | D

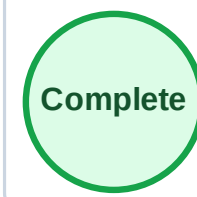
Visited: A, B, C, F, H, I, L, N



DFS Traversal

Step 19: TARGET NODE K REACHED! Path (4 hops): A → B → C → N → K

Legend



Stack (top at right): O | M | J | G | E | D

Visited: A, B, C, F, H, I, K, L, N

